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Centered wave bubbles with sonic boundary of pseudosteady Guderley Mach reflection configurations in gas dynamics

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ABSTRACT

In an attempt to resolve the von Neumann triple point paradox in shock reflection phenomenon, a new type of reflection configuration, called Guderley Mach reflection, was observed both in numerical simulations and physical experiments recently. In this type of reflection configuration there is a sequence of triple points, with a centered expansion fan at each triple point. This paper studies the existence of these centered expansion fans. In order to construct such a centered expansion fan, the problem of centered wave bubble with sonic boundary is introduced. Existence of global classical solution to the centered wave bubble problem for two-dimensional 2D isentropic irrotational pseudosteady Euler equations is obtained constructively. The techniques used here can be useful for constructing centered wave flow patterns of other flow problems and for constructing global classical solutions to centered wave problems for some other types of quasilinear hyperbolic systems.

RÉSUMÉ

Dans une tentative de résolution du paradoxe du point triple de Von Neumann dans le phénomène de réflexion de choc, un nouveau type de configuration de réflexion, appelée réflexion de Guderley Mach, a été observée récemment à la fois dans les simulations numériques et les expériences physiques. Dans ce type de configuration de réflexion, il y a une suite de points triples avec un éventail d'expansion centré à chaque point triple. Dans cet article on étudie l'existence de ces éventails d'expansion centrés. Pour cela, le problème de bulles d'ondes centrées avec un bord sonique est introduit. L'existence globale de solutions classiques au problème d'onde centrée pour les équations d'Euler isentropiques irrotationnelles pseudo-stationnaires en 2D est obtenue de manière constructive. Les techniques utilisées ici peuvent être utiles dans la construction de modèles de flux d'ondes centrées d'autres problèmes d'écoulement et de construction des solutions classiques globales aux problèmes d'ondes centrées de certains autres types de systèmes hyperboliques quasi-linéaires.

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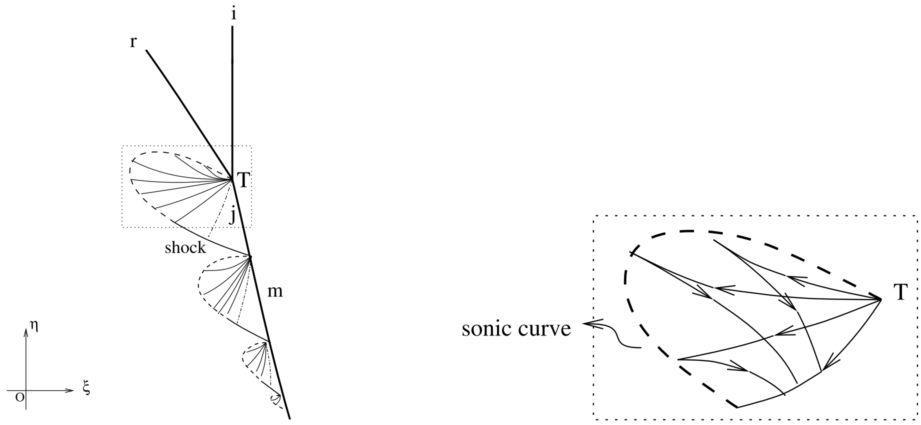

Fig. 1. Left: a schematic illustration of GMR configuration on the self-similar (ξ, η) plane based on numerical results of Tesdall et al. [20–22]; right: a centered wave bubble with sonic boundary.

1. Introduction

When a planar shock moving with a constant speed hits a sharp straight wedge, a shock reflection phenomenon will occur. Depending on the strength of the incident shock and the wedge angle, different reflection wave configurations can be obtained. Shock wave reflection phenomena were first reported by Ernst Mach [15] in 1878. And a systematic development of the theory, known as the two and three shock local theory was initiated by von Neumann and his collaborators [16] in 1943. Experiments and numerical simulations have shown that various patterns of shock reflection may occur, including the regular and Mach reflections (cf. Ben-Dor [1]).

In the theory of shock reflection there is a longstanding puzzle, known as the von Neumann triple point paradox. It refers to the existence of reflection configuration which looks similar to Mach reflection, but von Neumann's three shock theory indicates that there is not a solution.

In order to resolve the paradox, Guderley [3] proposed a modified Mach reflection. He added a centered expansion fan behind the triple point. He also demonstrated that the local structures consisting of three shocks, a centered expansion fan, and a contact discontinuity were possible (cf. Coruant and Friedrichs [2]).

For a long time this pattern of reflection was not observed either in physical experiments or in numerical simulations. Until recently, Hunter et al. [5,24], and Vasil'ev and Kraiko [23] found numerical evidence of the existence of a centered wave and a supersonic patch behind the triple point by investigating numerically weak shock reflection. Subsequently, Tesdall et al. [20–22] (see Fig. 1) found that there exists a weak shock at the edge of the supersonic patch and the interaction of the weak shock with the Mach shock generates a new triple point and a tiny supersonic patch. Hence a quite complex flow structure, in which there is a sequence of triple points and tiny supersonic patches and a weak shock at the edge of each supersonic patch. In addition, they conjectured that the sequence of triple points and supersonic patches was infinite in an inviscid weak shock reflection. This new type of reflection configuration is now referred to as Guderley Mach reflection (GMR) which provides a solution of the triple point paradox. Recently, GMR was also observed in physical experiments (cf. Skews et al. [18,19]).

There is still a long way to go to solve the triple point paradox theoretically. As far as we know, even existence of solution to Guderley Mach reflection configuration seems to be still an open problem. In this paper, we study the existence of these centered expansion fans. In order to construct such a centered expansion fan, we consider a centered wave bubble problem. Existence of global classical solution to the centered wave bubble problem for 2D isentropic irrotational pseudosteady Euler equations is obtained constructively.

This paper is organized as follows. In Section 2 we present our basic systems, the characteristic directions, the characteristic angles, the pseudo-Mach angle, and a series of characteristic equations in terms of the characteristic angles and the speed of sound. In Section 3 we give the centered wave bubble problem and our main result. We also consider a trivial case of the centered wave problem to which the solution is just a centered simple wave. Section 4 is devoted to prove the existence of global C^1 solution to the centered wave problem. As far as we know, there are few results about existence of global classical solutions to centered wave problems for hyperbolic systems. The techniques used here may be also useful for constructing global classical solutions to centered wave problems for some other quasilinear hyperbolic systems.

2. Systems of 2D isentropic irrotational pseudosteady flow

2.1. 2D isentropic irrotational pseudosteady Euler equations

We consider the 2D isentropic pseudosteady Euler equations

$$\begin{cases} (\rho U)_\xi + (\rho V)_\eta + 2\rho = 0, \\ UU_\xi + VU_\eta + \left(\dfrac{c^2}{\gamma-1}\right)_\xi + U = 0, \\ UV_\xi + VV_\eta + \left(\dfrac{c^2}{\gamma-1}\right)_\eta + V = 0, \end{cases} \tag{2.1}$$

where $(\xi = \frac{x}{t}, \eta = \frac{y}{t}), (U, V) = (u - \xi, v - \eta)$ is called the pseudo-velocity, (u, v) is the velocity, p is the pressure, ρ is the density, and $c = \sqrt{p'(\rho)}$ is the sound speed (cf. Zheng [26]). For polytropic gases, we have $p = A\rho^\gamma$, in which the coefficient A is a constant and the adiabatic exponent γ is a constant between 1 and 3. If the flow is also irrotational, i.e.,

$$u_\eta - v_\xi = 0, \tag{2.2}$$

we can introduce a potential function $\varphi(\xi, \eta)$, such that $\varphi_\xi = U$ and $\varphi_\eta = V$. Then by the last two equations of system (2.1) we have the pseudo-Bernoulli law

$$\frac{q^2}{2} + \frac{c^2}{\gamma-1} + \varphi = \text{Const.}, \tag{2.3}$$

where $q^2 = U^2 + V^2$. The first equation of (2.1) can by (2.3) be written in the form

$$(c^2 - U^2)u_\xi - UV(u_\eta + v_\xi) + (c^2 - V^2)v_\eta = 0. \tag{2.4}$$

The eigenvalues of Eqs. (2.2), (2.4) are determined by

$$(V - \lambda U)^2 - c^2(1 + \lambda^2) = 0, \tag{2.5}$$

which yields

$$\lambda = \lambda_\pm = \frac{UV \pm c\sqrt{U^2 + V^2 - c^2}}{U^2 - c^2}. \tag{2.6}$$

So, if and only if $U^2 + V^2 > c^2$, Eqs. (2.2), (2.4) are hyperbolic and have two families of wave characteristics defined by

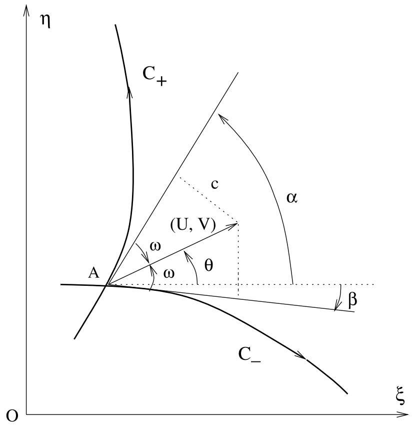

Fig. 2. Characteristic directions and characteristic angles.

$$C_\pm : \ \frac{\mathrm{d}\eta}{\mathrm{d}\xi} = \lambda_\pm \tag{2.7}$$

Eqs. (2.2), (2.4) can be written in the characteristic form

$$\begin{cases} \partial_+ u + \lambda_- \partial_+ v = 0, \\ \partial_- u + \lambda_+ \partial_- v = 0, \end{cases} \tag{2.8}$$

in which $\partial_\pm = \partial_\xi + \lambda_\pm \partial_\eta$.

2.2. Characteristics

From (2.5) we have $c = \left|(U,V)\cdot\left(\frac{\lambda}{\sqrt{1+\lambda^2}}, -\frac{1}{\sqrt{1+\lambda^2}}\right)\right|$, which means that the component of the pseudo-velocity normal to the direction of a C characteristic is equal to the speed of sound. Equivalently, it can be stated as that the tangent line of a C characteristic at a point is tangent to the sonic circle of the state at the point. Furthermore, a C characteristic must be straight if (u, v, c) is constant along it.

The direction of the $C_\pm$ characteristics is defined as the tangent direction that forms an acute angle ω with the pseudo-velocity vector (U, V). By computation, we have that the C_+ characteristic forms with the pseudoflow direction the angle ω from C_+ to (U, V) in clockwise direction, the C_- characteristic forms with the pseudoflow direction the angle ω from C_- to (U, V) in counterclockwise direction, and $\sin^2\omega = \frac{c^2}{q^2}$. The angle ω is called the pseudo-Mach angle. For pseudosonic flow, we have $\omega = \pi/2$. See Fig. 2.

Following Li and Zheng [8], we introduce the concept of characteristic angles. The C_+ (C_-) characteristic angle is defined as the angle between the C_+ (C_-) characteristic direction and the positive ξ-axis. We denote respectively by α and β the C_+ and C_- characteristic angles, where α and β satisfy $0 < \alpha - \beta < \pi$. Let θ be the angle between the direction of pseudo-velocity and the positive ξ-axis. We have

$$\theta = \frac{\alpha+\beta}{2}, \quad \omega = \frac{\alpha-\beta}{2}, \quad \alpha = \theta + \omega, \quad \beta = \theta - \omega. \tag{2.9}$$

Therefore, u, v, and c can be expressed in terms of α, β, and c by

$$u - \xi = c\frac{\cos\theta}{\sin\omega}, \quad v - \eta = c\frac{\sin\theta}{\sin\omega}. \tag{2.10}$$

In terms of the variables α, β, and c, the pseudo-Bernoulli law (2.3) can be written as

$$\frac{c^2}{2\sin^2\omega} + \frac{c^2}{\gamma-1} + \varphi = \mathrm{Const.} \tag{2.11}$$

2.3. Characteristic equations in terms of α, β, and c

In this part, we shall cite a series of characteristic equations in terms of the variables α, β, and c from Section 2 of Li, Yang, and Zheng [9].

In the following discussion, we define

$$\bar{\partial}_+ = \cos\alpha\partial_\xi + \sin\alpha\partial_\eta \quad \text{and} \quad \bar{\partial}_- = \cos\beta\partial_\xi + \sin\beta\partial_\eta. \tag{2.12}$$

By (2.10) we have

$$\bar{\partial}_\pm u = \cos(\theta\pm\omega) + \frac{\cos\theta}{\sin\omega}\bar{\partial}_\pm c + \frac{c\cos\alpha\bar{\partial}_\pm\beta - c\cos\beta\bar{\partial}_\pm\alpha}{2\sin^2\omega}, \tag{2.13}$$

$$\bar{\partial}_\pm v = \sin(\theta\pm\omega) + \frac{\sin\theta}{\sin\omega}\bar{\partial}_\pm c + \frac{c\sin\alpha\bar{\partial}_\pm\beta - c\sin\beta\bar{\partial}_\pm\alpha}{2\sin^2\omega}. \tag{2.14}$$

Inserting (2.13) and (2.14) into (2.8), we get

$$\bar{\partial}_+ c = -\frac{\cos 2\omega}{\cot\omega} + \frac{c}{\sin 2\omega}(\bar{\partial}_+\alpha - \cos 2\omega\bar{\partial}_+\beta) \tag{2.15}$$

and

$$\bar{\partial}_- c = -\frac{\cos 2\omega}{\cot\omega} + \frac{c}{\sin 2\omega}(\cos 2\omega\bar{\partial}_-\alpha - \bar{\partial}_-\beta). \tag{2.16}$$

Differentiating the pseudo-Bernoulli law (2.11) and using (2.13) and (2.14), we have

$$\left(\frac{1}{\sin^2\omega} + \frac{1}{\kappa}\right)\bar{\partial}_\pm c = \frac{c\cos\omega}{2\sin^3\omega}(\bar{\partial}_\pm\alpha - \bar{\partial}_\pm\beta) - \cot\omega, \tag{2.17}$$

where $\kappa = \frac{\gamma-1}{2}$. Thus, we have the following closed system in terms of α, β, and φ:

$$\begin{cases} c\bar{\partial}_+\alpha = \Omega\cos^2\omega(2\sin^2\omega + c\bar{\partial}_+\beta), \\ c\bar{\partial}_-\beta = \Omega\cos^2\omega(-2\sin^2\omega + c\bar{\partial}_-\alpha), \\ \bar{\partial}_\perp\varphi = 0. \end{cases} \tag{2.18}$$

In system (2.18), the first two equations are derived from (2.15)–(2.17), $\bar{\partial}_\perp = -\sin\theta\partial_\xi + \cos\theta\partial_\eta$, $\Omega = \frac{3-\gamma}{\gamma+1} - \tan^2\omega$, and the variable $c = c(\omega,\varphi) := \sqrt{-\frac{2(\gamma-1)\varphi\sin^2\omega}{(\gamma-1)+2\sin^2\omega}}$ by letting the constant of the pseudo-Bernoulli law (2.11) be 0.

From (2.15), (2.16), and the first two equations of system (2.18) we also have

$$c\bar{\partial}_+\alpha = -\frac{\sin 2\omega}{2\mu^2}\Omega\bar{\partial}_+ c, \tag{2.19}$$

$$c\bar{\partial}_+\beta = -\frac{\tan\omega}{\mu^2}\bar{\partial}_+ c - 2\sin^2\omega, \tag{2.20}$$

$$c\bar{\partial}_-\beta = \frac{\sin 2\omega}{2\mu^2}\Omega\bar{\partial}_- c, \tag{2.21}$$

$$c\bar{\partial}_-\alpha = \frac{\tan\omega}{\mu^2}\bar{\partial}_- c + 2\sin^2\omega, \tag{2.22}$$

where $\mu^2 = \frac{\gamma-1}{\gamma+1}$. From (2.13), (2.14), and (2.19)–(2.22) we have

$$\bar\partial_\pm u = \pm\kappa^{-1}\sin(\theta\mp\omega)\bar\partial_\pm c, \quad \bar\partial_\pm v = \mp\kappa^{-1}\cos(\theta\mp\omega)\bar\partial_\pm c. \tag{2.23}$$

System (2.18) can be also written in the following diagonal form

$$\begin{cases} \bar\partial_+ r_- = h(c,\omega), \\ \bar\partial_- r_+ = h(c,\omega), \\ \bar\partial_\perp \varphi = 0, \end{cases} \tag{2.24}$$

where

$$r_\pm = r_\pm(\alpha,\beta) = \frac{1}{\mu}\arctan(\mu\cot\omega) \pm (\theta\pm\omega), \quad h(c,\omega) = -\frac{\sin^2\omega(\cos 2\omega-\kappa)}{c(\kappa+\sin^2\omega)}. \tag{2.25}$$

Proposition 2.1. *Along the curve $r_+ = Const.$ there holds $\frac{d\alpha}{d\omega} > 0$, and along the curve $r_- = Const.$ there holds $\frac{d\beta}{d\omega} < 0$.*

Proof. This proposition can be proved by direct computation, we omit the details. □

2.4. Characteristic decompositions for the variables α, β, and c

In this part, we shall present various characteristic decompositions for the variables α, β, and c. We first cite the following commutator relation from Li, Zhang, and Zheng [7].

Proposition 2.2 *(Commutator relation). For any quantity $I(\xi,\eta)$ we have*

$$\bar\partial_-\bar\partial_+ I - \bar\partial_+\bar\partial_- I = \frac{1}{\sin 2\omega}\big[(\cos 2\omega\bar\partial_+\beta - \bar\partial_-\alpha)\bar\partial_- I - (\bar\partial_+\beta - \cos 2\omega\bar\partial_-\alpha)\bar\partial_+ I\big]. \tag{2.26}$$

Proposition 2.3. *For the variable c, we have the following characteristic decompositions*

$$\begin{cases} c\bar\partial_+\bar\partial_- c = \bar\partial_- c\left(\sin 2\omega + \dfrac{1}{2\mu^2\cos^2\omega}\bar\partial_- c + \left(\dfrac{\Omega\cos 2\omega}{2\mu^2}+1\right)\bar\partial_+ c\right), \\ c\bar\partial_-\bar\partial_+ c = \bar\partial_+ c\left(\sin 2\omega + \dfrac{1}{2\mu^2\cos^2\omega}\bar\partial_+ c + \left(\dfrac{\Omega\cos 2\omega}{2\mu^2}+1\right)\bar\partial_- c\right). \end{cases} \tag{2.27}$$

Proof. This proposition was due to Li, Yang, and Zheng [9]. For the sake of completeness, we sketch the proof. It follows from (2.23) and (2.26) that

$$\begin{aligned} \bar\partial_-\bar\partial_+ u - \bar\partial_+\bar\partial_- u &= \bar\partial_+\left[\kappa^{-1}\sin\alpha\bar\partial_- c\right] + \bar\partial_-\left[\kappa^{-1}\sin\beta\bar\partial_+ c\right] \\ &= -\frac{1}{\sin 2\omega}\big[(\bar\partial_+\beta - \cos 2\omega\bar\partial_-\alpha)\kappa^{-1}\sin\beta\bar\partial_+ c - (\bar\partial_-\alpha - \cos 2\omega\bar\partial_+\beta)\kappa^{-1}\sin\alpha\bar\partial_- c\big]. \end{aligned} \tag{2.28}$$

Hence, we have

$$\begin{aligned} \sin\alpha\bar\partial_+\bar\partial_- c + \sin\beta\bar\partial_-\bar\partial_+ c = -\frac{1}{\sin 2\omega}\Big(&(\sin\beta\bar\partial_+\beta - \sin\beta\cos 2\omega\bar\partial_-\alpha + \cos\beta\sin 2\omega\bar\partial_-\beta)\bar\partial_+ c \\ &- (\sin\alpha\bar\partial_-\alpha - \sin\alpha\cos 2\omega\bar\partial_+\beta - \cos\alpha\sin 2\omega\bar\partial_+\alpha)\bar\partial_- c\Big). \end{aligned} \tag{2.29}$$

Applying the commutator relation (2.26) for c, we get

$$\bar{\partial}_-\bar{\partial}_+c - \bar{\partial}_+\bar{\partial}_-c = \frac{1}{\sin 2\omega}\Big((\cos 2\omega\bar{\partial}_+\beta - \bar{\partial}_-\alpha)\bar{\partial}_-c - (\bar{\partial}_+\beta - \cos 2\omega\bar{\partial}_-\alpha)\bar{\partial}_+c\Big). \tag{2.30}$$

Inserting (2.30) into (2.29), we get

$$\begin{aligned}(\sin\alpha + \sin\beta)\bar{\partial}_+\bar{\partial}_-c = -\frac{1}{\sin 2\omega}\Big(&\cos\beta\sin 2\omega\bar{\partial}_-\beta\bar{\partial}_+c + (\sin\alpha+\sin\beta)\cos 2\omega\bar{\partial}_+\beta\bar{\partial}_-c \\ &- (\sin\alpha+\sin\beta)\bar{\partial}_-\alpha\bar{\partial}_-c + \cos\alpha\sin 2\omega\bar{\partial}_+\alpha\bar{\partial}_-c\Big).\end{aligned} \tag{2.31}$$

Thus, by (2.19)–(2.22) we get the first equation of (2.27). The proof for the other is similar. We then have this proposition. □

Proposition 2.4. *For the variables α and β, we have*

$$\begin{cases} c\bar{\partial}_+\bar{\partial}_-\alpha = M_1\bar{\partial}_-\alpha - \dfrac{\sin 2\omega}{2}(3\tan^2\omega - 1)\bar{\partial}_+\alpha \\ c\bar{\partial}_-\bar{\partial}_+\beta = M_2\bar{\partial}_+\beta - \dfrac{\sin 2\omega}{2}(3\tan^2\omega - 1)\bar{\partial}_-\beta, \end{cases} \tag{2.32}$$

where

$$M_1 = \frac{1}{\sin 2\omega}\left(4\sin^4\omega(-1+\Omega\cos^2\omega) + c\bar{\partial}_-\alpha - \left(1 - \frac{\Omega\sin^2 2\omega}{2}\right)c\bar{\partial}_+\beta\right), \tag{2.33}$$

$$M_2 = \frac{1}{\sin 2\omega}\left(4\sin^4\omega(-1+\Omega\cos^2\omega) - c\bar{\partial}_+\beta + \left(1 - \frac{\Omega\sin^2 2\omega}{2}\right)c\bar{\partial}_-\alpha\right). \tag{2.34}$$

Proof. This proposition was also due to Li, Yang, and Zheng [9]. This proposition can be obtained by directly inserting (2.20) and (2.22) into (2.27). □

Proposition 2.5 *(The second order equation of c in homogeneous form).*

$$\begin{cases} c\bar{\partial}_+\left(\dfrac{\bar{\partial}_-c}{\sin^2\omega}\right) = \dfrac{1}{2\mu^2\cos^2\omega}\dfrac{\bar{\partial}_-c}{\sin^2\omega}\left(\bar{\partial}_+c + \bar{\partial}_-c\right) - \left(\dfrac{8\sin^2\omega}{\gamma-1} + 2\right)\dfrac{\bar{\partial}_+c\bar{\partial}_-c}{\sin^2\omega}, \\ c\bar{\partial}_-\left(\dfrac{\bar{\partial}_+c}{\sin^2\omega}\right) = \dfrac{1}{2\mu^2\cos^2\omega}\dfrac{\bar{\partial}_+c}{\sin^2\omega}\left(\bar{\partial}_-c + \bar{\partial}_+c\right) - \left(\dfrac{8\sin^2\omega}{\gamma-1} + 2\right)\dfrac{\bar{\partial}_+c\bar{\partial}_-c}{\sin^2\omega}. \end{cases} \tag{2.35}$$

Proof. This proposition was due to Li and Zheng [10]. □

3. The centered wave bubble problem

3.1. Isentropic irrotational pseudosteady centered waves

We first present the centered waves to Eqs. (2.18).

Definition 1. Let $Y(t)$ be an angular domain with curved boundaries:

$$Y(t) := \{(\xi,\eta) \mid 0 \le \xi \le t,\ \eta_1(\xi) \le \eta \le \eta_2(\xi)\}, \tag{3.1}$$

where $\eta_1(0) = \eta_2(0) = 0$ and $\zeta_1 = \eta_1'(0) < \eta_2'(0) = \zeta_2$. A function $(\alpha,\beta,\varphi)(\xi,\eta)$ is called a C_+ type centered wave solution for system (2.18) with the origin $(0,0)$ as the center point if the following properties are satisfied (cf. Li and Yu [12], pp. 188–190) (see Fig. 3):

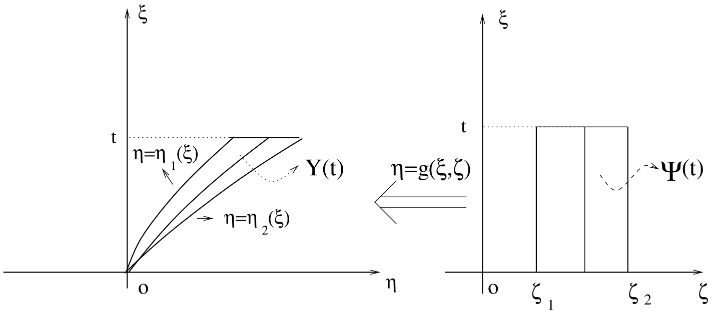

Fig. 3. A centered wave flow region in the (ξ,η) plane (left); a centered wave flow region in the (ξ,ζ) plane (right).

1. (α,β,φ) can be implicitly determined by the functions $\eta = g(\xi,\zeta)$ and $(\alpha,\beta,\varphi)(\xi,\eta) = (\hat\alpha,\hat\beta,\hat\varphi)(\xi,\zeta)$ defined on a rectangular domain $\Psi(t) := \{(\xi,\zeta) \mid 0 \le \xi \le t,\ \ \zeta_1 \le \zeta \le \zeta_2\}$. Moreover, g and $(\hat\alpha,\hat\beta,\hat\varphi)$ belong to $C^1(\Psi(t))$, and for any $(\xi,\zeta) \in \Psi(t) \setminus \{\xi = 0\}$ there holds $g_\zeta(\xi,\zeta) > 0$.
2. The function $(\alpha,\beta,\varphi)(\xi,\eta)$ defined above satisfies Eqs. (2.18) on $Y(t) \setminus (0,0)$.
3. For any $\zeta \in [\zeta_1,\zeta_2]$, $\eta = g(\xi,\zeta)$ gives the C_+ characteristic line passing through the origin $(0,0)$ with the slope ζ at the origin, i.e.

$$g_\xi(\xi,\zeta) = \tan\hat\alpha(\xi,\zeta), \quad g(0,\zeta) = 0, \quad g_\xi(0,\zeta) = \zeta. \tag{3.2}$$

4. $\zeta = \zeta_1$ and $\zeta = \zeta_2$ correspond to $\eta = \eta_1(\xi)$ and $\eta = \eta_2(\xi)$, respectively.

$(\hat\alpha,\hat\beta,\hat\varphi)(0,\zeta)$ is called the principal part of this C_+ type centered wave and ζ the characteristic parameter. Similarly, we can define C_- type centered wave.

Since

$$\frac{\partial}{\partial\xi} = \frac{\partial}{\partial\xi} + \frac{\partial g}{\partial\xi}\frac{\partial}{\partial\eta}, \qquad \frac{\partial}{\partial\zeta} = \frac{\partial g}{\partial\zeta}\frac{\partial}{\partial\eta}, \tag{3.3}$$

by (2.24) we have

$$\begin{cases} \cos\hat\alpha\dfrac{\partial\hat r_-}{\partial\xi} = h(\hat c,\ \hat\omega), \\ \cos\hat\beta\dfrac{\partial g}{\partial\zeta}\dfrac{\partial\hat r_+}{\partial\xi} - \dfrac{\sin 2\hat\omega}{\cos\hat\alpha}\dfrac{\partial\hat r_+}{\partial\zeta} = \dfrac{\partial g}{\partial\zeta}h(\hat c,\ \hat\omega) \\ -\sin\hat\theta\dfrac{\partial g}{\partial\zeta}\dfrac{\partial\hat\varphi}{\partial\xi} + (\cos\hat\theta - \sin\hat\theta\tan\hat\alpha)\dfrac{\partial\hat\varphi}{\partial\zeta} = 0. \end{cases} \tag{3.4}$$

From (3.2) we have

$$\frac{\partial g(\xi,\zeta)}{\partial\zeta} = \int_0^\xi \sec^2\hat\alpha(\tau,\zeta)\frac{\partial\hat\alpha(\tau,\zeta)}{\partial\zeta}d\tau. \tag{3.5}$$

Hence, by letting $\xi \to 0$ we have the principle part of the C_+ type centered wave satisfies

$$\tan\hat\alpha(0,\zeta) = \zeta, \quad \hat r_+(0,\zeta) = \text{Const.}, \qquad \text{and} \ \ \hat\varphi(0,\zeta) = \text{Const.} \tag{3.6}$$

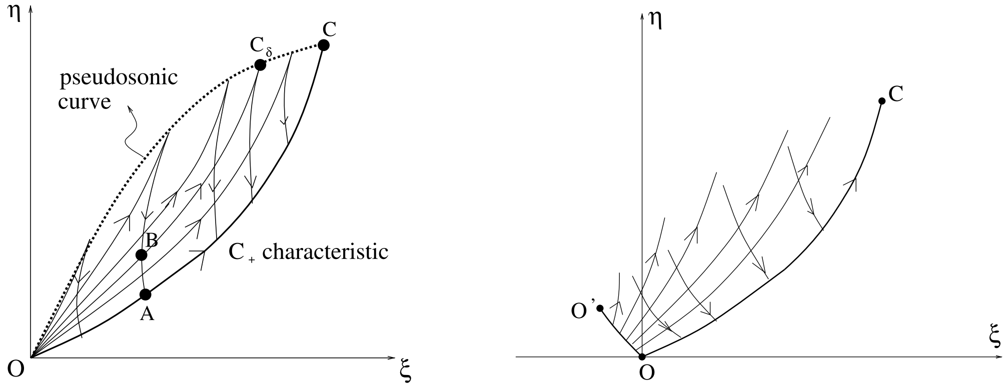

Fig. 4. The centered wave bubble problem; right: Goursat problem.

3.2. The centered wave bubble problem

In order to construct a centered wave bubble with sonic boundary of pseudosteady Guderley Mach reflection configuration, we consider the following centered wave problem.

Referring to Fig. 4 (left), let $\widetilde{OC}$ be a smooth curve passing through the origin O. We prescribe

$$(\alpha, \beta, \varphi) = \left(\alpha^b, \beta^b, \varphi^b\right)(\xi, \eta) \quad \text{on } \widetilde{OC}, \tag{3.7}$$

so that $\widetilde{OC}$ is a C_+ characteristic curve with the direction of O points to C, the flow at the point C is just pseudosonic, and the first equation of (2.18) holds along $\widetilde{OC}$. We also prescribe the principle part

$$\left(\hat{\alpha}, \hat{\beta}, \hat{\varphi}\right)(0, \zeta) = \left(\hat{\alpha}^p, \hat{\beta}^p, \hat{\varphi}^p\right)(\zeta) \qquad \text{as} \quad 0 < \zeta_1 \le \zeta \le \zeta_2, \tag{3.8}$$

so that

$$\begin{aligned} &\left(\hat{\alpha}^p, \hat{\beta}^p, \hat{\varphi}^p\right)(\zeta_1) = \left(\alpha^b, \beta^b, \varphi^b\right)(0,0), \quad \hat{\alpha}^p(\zeta_2) - \hat{\beta}^p(\zeta_2) = \pi, \quad \tan\hat{\alpha}^p(\zeta) = \zeta, \\ &\hat{\varphi}^p(\zeta) \equiv \varphi^b(0,0), \quad \text{and} \quad \hat{r}_+^p(\zeta) := r_+(\hat{\alpha}^p, \hat{\beta}^p) \equiv r_+(\alpha^b(0,0), \beta^b(0,0)) \quad \text{as} \quad \zeta \in [\zeta_1, \zeta_2]. \end{aligned} \tag{3.9}$$

Then find a C_+ type centered wave for system (2.18), such that it satisfies (3.7) and its principle part satisfies (3.8).

Remark 3.1. This type of centered wave problem looks like a Goursat-type boundary value problem if we "unfold" the center point; see Fig. 4 (right).

3.3. Simple wave solution to the centered wave bubble problem

In order to make the centered wave problem (2.18), (3.7), and (3.8) easy to understand, we consider a trivial case.

Theorem 3.2. *If the boundary condition (3.7) satisfies*

$$\alpha^b \equiv \alpha_0 \quad \textit{and} \quad c^b \equiv c_0, \tag{3.10}$$

where α_0 and c_0 are two constants (that is to say $\widetilde{OC}$ is a straight C_+ characteristic with the direction of O points to C), then the centered wave problem (2.18), (3.7), and (3.8) admits a centered simple wave solution. See Fig. 5.

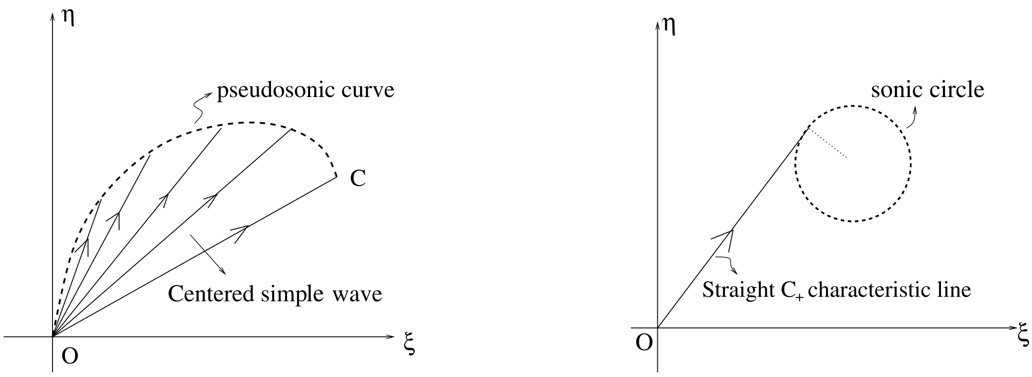

Fig. 5. Centered simple wave solution to the centered wave problem.

Proof. Let $(u,v,c)(\xi,\eta) := (\hat{u},\hat{v},\hat{c})(\zeta)$, where $\zeta := \frac{\eta}{\xi}$, and the functions $\hat{u}(\zeta)$, $\hat{v}(\zeta)$, and $\hat{c}(\zeta)$ are determined by

$$\begin{cases} \hat{u}(\zeta) = \dfrac{\hat{c}(\zeta)\cos\hat{\theta}(\zeta)}{\sin\hat{\omega}(\zeta)}, \quad \hat{v}(\zeta) = \dfrac{\hat{c}(\zeta)\sin\hat{\theta}(\zeta)}{\sin\hat{\omega}(\zeta)}, \quad \tan\hat{\alpha}(\zeta) = \zeta, \quad \hat{\theta}(\zeta) = \hat{\alpha}(\zeta) - \hat{\omega}(\zeta), \\ \dfrac{\hat{c}^2(\zeta)}{2\sin^2\hat{\omega}(\zeta)} + \dfrac{\hat{c}^2(\zeta)}{\gamma-1} = \dfrac{c_0^2}{2\sin^2\omega^b(0,0)} + \dfrac{c_0^2}{\gamma-1}, \quad 2\omega^b(0,0) = \alpha_0 - \beta^b(0,0), \\ \arctan\big(\mu\cot\hat{\omega}(\zeta)\big) + \mu\hat{\alpha}(\zeta) \equiv \arctan\big(\mu\cot\omega^b(0,0)\big) + \mu\alpha_0. \end{cases} \tag{3.11}$$

We shall show that $(u,v,c)(\xi,\eta)$ is actually a centered simple wave solution of problem (2.18), (3.7), and (3.8).

In fact, from (3.11) we have

$$\frac{\hat{u}(\zeta)\hat{v}(\zeta) + \hat{c}(\zeta)\sqrt{\hat{u}^2(\zeta)+\hat{v}^2(\zeta)-\hat{c}^2(\zeta)}}{\hat{u}^2(\zeta)-\hat{c}^2(\zeta)} = \zeta. \tag{3.12}$$

Then, for each fixed ζ, the straight ray $\eta = \zeta\xi$ is tangent to the sonic circle $\big(\hat{u}(\zeta)-\xi\big)^2 + \big(\hat{v}(\zeta)-\eta\big)^2 = \hat{c}^2(\zeta)$, and hence $\alpha(\xi,\eta) = \hat{\alpha}(\eta/\xi) = \arctan\frac{\eta}{\xi}$; see Fig. 5 (right). Therefore, we have $\bar{\partial}_+ u = \bar{\partial}_+ v = \bar{\partial}_+ c = 0$ and accordingly $\bar{\partial}_+ u + \lambda_- \bar{\partial}_+ v = 0$. By computation, we have

$$\bar{\partial}_- u + \lambda_+ \bar{\partial}_- v = \big(\hat{u}'(\zeta) + \lambda_+ \hat{v}'(\zeta)\big)\bar{\partial}_-\zeta = \frac{1}{\cos\hat{\alpha}}\left(\frac{\hat{c}'\cos\hat{\omega}}{\sin\hat{\omega}} + \hat{c}\hat{\alpha}' - \frac{\hat{c}\hat{\omega}'}{\sin^2\hat{\omega}}\right)\bar{\partial}_-\zeta = 0, \tag{3.13}$$

since $\hat{\alpha}' = \frac{\hat{\omega}'}{\sin^2\hat{\omega}+\mu^2\cos^2\hat{\omega}}$ and $\frac{\hat{c}'}{\sin^2\hat{\omega}} - \frac{\hat{c}\cos\hat{\omega}\hat{\omega}'}{\sin^3\hat{\omega}} + \frac{2\hat{c}'}{\gamma-1} = 0$ from (3.11). By computation, we have

$$\bar{\partial}_+\left(\frac{1}{2}(U^2+V^2) + \frac{c^2}{\gamma-1} + \varphi\right) = (u-\xi)\bar{\partial}_+ u + (v-\eta)\bar{\partial}_+ v + \frac{2c}{\gamma-1}\bar{\partial}_+ c = 0 \tag{3.14}$$

and

$$\begin{aligned} \bar{\partial}_-\left(\frac{1}{2}(U^2+V^2) + \frac{c^2}{\gamma-1} + \varphi\right) &= (u-\xi)\bar{\partial}_- u + (v-\eta)\bar{\partial}_- v + \frac{2c}{\gamma-1}\bar{\partial}_- c \\ &= \left(\hat{u}(\zeta)\hat{u}'(\zeta) + \hat{v}(\zeta)\hat{v}'(\zeta) + \frac{2}{\gamma-1}\hat{c}(\zeta)\hat{c}'(\zeta) - \xi\hat{u}'(\zeta) - \eta\hat{v}'(\zeta)\right)\bar{\partial}_-\zeta \\ &= \left(-\xi\hat{u}'(\zeta) - \eta\hat{v}'(\zeta)\right)\bar{\partial}_-\zeta = -\xi\left(\hat{u}'(\zeta) + \zeta\hat{v}'(\zeta)\right)\bar{\partial}_-\zeta = 0. \end{aligned} \tag{3.15}$$

We then have this theorem. □

3.4. Main theorem

The main result of this paper is the following theorem:

Theorem 3.3 *(Main theorem). Assume that* $(\alpha^b, \beta^b, c^b) \in C^1(\widetilde{OC})$, $\omega^b(0,0) > \max\left\{\arctan\sqrt{\frac{3-\gamma}{1+\gamma}}, \arctan\frac{1}{3}\right\}$, *and* $\bar{\partial}_+ c^b\mid_{\widetilde{OC}} > 0$, *then the centered wave problem (2.18), (3.7), and (3.8) admits a* C^1 *solution on a domain closed by* $\widetilde{OC}$ *and a pseudosonic curve connecting* O *and* C.

4. Existence of global classical solution to the centered wave bubble problem

In what follows we shall take the following three steps (see Fig. 4 (left)):

1. Find a local C^1 centered wave solution with small amplitude. Precisely speaking, for sufficiently small δ, find a solution on the domain closed $\widetilde{OC}$, a C_+ characteristic curve $\widetilde{OB}$ passing through the origin and with the slope $\zeta_1+\delta$ at the origin, and a C_- characteristic curve $\widetilde{BA}$ passing through some point A on $\widetilde{OC}$;
2. By solving a Goursat problem for system (2.18) with $\widetilde{BA}$ and $\widetilde{AC}$ as the characteristic boundaries, find the flow in a domain closed by $\widetilde{BA}$, $\widetilde{AC}$, a C_+ characteristic curve $\widetilde{BC_\delta}$ passing through the point B, and a pseudosonic curve connecting C and C_δ;
3. Establish C^1 norm estimate of solution on the characteristic curve $\widetilde{OC_\delta}$ and repeat the first and second steps.

4.1. Local C^1 solution to the centered wave bubble problem

Centered wave problems for general first order quasilinear hyperbolic systems were first proposed and studied by Li and Yu [12–14]. They obtained local centered wave solutions with small amplitude for general first order quasilinear hyperbolic systems; see Theorem 7.1 on page 210 of Li and Yu [12]. Subsequently, Zhou [27,28] obtained local centered wave solutions with large amplitude for general first order quasilinear hyperbolic systems. These results were obtained under the assumption that boundary conditions are at least C^2 smooth. In this paper, we prove existence of global C^1 solution to the centered wave problem for only C^1 smooth boundary data.

Since system (2.1) is invariant under rotation of coordinates, we assume that $\alpha^b(0,0) = \omega^b(0,0) = -\beta^b(0,0)$. Therefore, there exists a $\xi' > 0$, such that

$$\frac{\omega^b(0,0)}{2} \le \alpha^b\big(\xi, g(\xi,\zeta_1)\big) \le \frac{\pi}{4} + \frac{\omega^b(0,0)}{2} \quad \text{as } \xi \in [0,\xi']. \tag{4.1}$$

Thus, on $\{(\xi,\zeta_1) \mid 0 \le \xi \le \xi'\}$ we can define

$$\hat{\varphi} = \hat{\varphi}^b(\xi) := \varphi^b\big(\xi, g(\xi,\zeta_1)\big) \qquad \text{and} \qquad \hat{r}_+ = \hat{r}_+^b(\xi) := r_+\big(\alpha^b(\xi, g(\xi,\zeta_1)), \beta^b(\xi, g(\xi,\zeta_1))\big)\ . \tag{4.2}$$

And on $\{(0,\zeta) \mid \zeta \ge \zeta_1\}$ we have

$$\hat{r}_- = \hat{r}_-^p(\zeta) := r_-\big(\arctan\zeta, \hat{\beta}^p(\zeta)\big). \tag{4.3}$$

By (2.25) and (3.9) we have

$$(\hat{r}_-^p)' = -2(1-\mu^2)\cos^2\hat{\alpha}^p\cos^2\hat{\omega}^p \tag{4.4}$$

and

$$(\hat{r}_-^p)'' = 4(1-\mu^2)\cos^2\hat{\alpha}^p\cos\hat{\omega}^p\big[\cos\hat{\alpha}^p\cos\hat{\omega}^p\sin\hat{\alpha}^p + \frac{3}{2}\cos^2\hat{\alpha}^p\sin\hat{\omega}^p - (1-\mu^2)\cos^2\hat{\alpha}^p\sin\hat{\omega}^p\cos^2\hat{\omega}^p\big]. \tag{4.5}$$

In what follows, we are going to study existence of local solution to Eqs. (3.4) with boundary conditions (4.2) and (4.3). Let

$$R(\delta_0) := \big\{(r_+, r_-) \mid |r_+ - \hat{r}_+^b(0)| \le \delta_0, |r_- - \hat{r}_-^p(\zeta_1)| \le \delta_0\big\}, \tag{4.6}$$

then for sufficiently small $\delta_0 > 0$, δ_0 depends only on hyperbolicity number

$$\vartheta := \frac{\pi}{2} - \omega^b(0,0), \tag{4.7}$$

there is a neighborhood $U(\delta_0)$ of $\big(\alpha^b(0,0), \beta^b(0,0)\big)$, such that the map $r_\pm = r_\pm(\alpha, \beta)$ is a C^2 bijection from $U(\delta_0)$ to $R(\delta_0)$, and for any $(r_+, r_-) \in R(\delta_0)$ and $\varphi \in [\hat{\varphi}^b(0), \hat{\varphi}^b(\delta_0)]$,

$$c(\omega(r_+, r_-), \varphi) > \tfrac{c^b(0,0)}{2}, \qquad |\, \alpha(r_+, r_-) - \alpha^b(0,0) \,| < \tfrac{\vartheta}{4}, \qquad |\, \beta(r_+, r_-) - \beta^b(0,0) \,| < \tfrac{\vartheta}{4}. \tag{4.8}$$

Let the domain

$$\Lambda(\delta) := \{(\xi, \zeta) \mid \xi \ge 0,\ \zeta \ge \zeta_1,\ \xi + \zeta - \zeta_1 \le \delta\} \tag{4.9}$$

and the function set

$$\begin{aligned}\Sigma(\delta) := \Big\{(r_+, r_-, \varphi) \in C^2\big(\Lambda(\delta)\big) \,\Big|\, & (r_+, r_-)(\xi,\zeta) \in R(\delta_0) \text{ and } \varphi(\xi,\zeta) \in \\ & \big[\hat{\varphi}^b(0), \hat{\varphi}^b(\delta_0)\big] \text{ as } (\xi,\zeta) \in \Lambda(\delta),\ \ r_-(0,\zeta) = \hat{r}_-^p(\zeta),\ \ r_+(\xi,\zeta_1) = \hat{r}_+^b(\xi), \\ & \varphi(\xi,\zeta_1) = \hat{\varphi}^b(\xi),\ \|(Dr_+, Dr_-, D\varphi)\|_{0;\Lambda(\delta)} \le d,\ \|(D^2r_+, D^2r_-, D^2\varphi)\|_{0;\Lambda(\delta)} \le n\Big\},\end{aligned} \tag{4.10}$$

where d and n are positive constants which will be determined later. For all $(\tilde{r}_+, \tilde{r}_-, \tilde{\varphi}) \in \Sigma(\delta)$, the operator $\mathcal{L}$ is defined by letting $(\hat{r}_+, \hat{r}_-, \hat{\varphi}) = \mathcal{L}(\tilde{r}_+, \tilde{r}_-, \tilde{\varphi})$ be the unique solution to linear equations

$$\begin{cases} \cos\tilde{\alpha}\dfrac{\partial \hat{r}_-}{\partial \xi} = h(\tilde{c},\ \tilde{\omega}), \\ a\cos\tilde{\beta}\dfrac{\partial \hat{r}_+}{\partial \xi} - \dfrac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\dfrac{\partial \hat{r}_+}{\partial \zeta} = ah(\tilde{c},\ \tilde{\omega}), \\ -a\sin\tilde{\theta}\dfrac{\partial\hat{\varphi}}{\partial\xi} + (\cos\tilde{\theta} - \sin\tilde{\theta}\tan\tilde{\alpha})\dfrac{\partial\hat{\varphi}}{\partial\zeta} = 0 \end{cases} \tag{4.11}$$

with boundary conditions (4.2) and (4.3). In system (4.11), $\tilde{\alpha} = \alpha(\tilde{r}_+, \tilde{r}_-)$, $\tilde{\beta} = \beta(\tilde{r}_+, \tilde{r}_-)$, $\tilde{\theta} = (\tilde{\alpha} + \tilde{\beta})/2$, $\tilde{\omega} = (\tilde{\alpha} - \tilde{\beta})/2$, $\tilde{c} = c(\tilde{\omega}, \tilde{\varphi})$, and

$$a = a(\xi,\zeta) = \int_0^\xi \sec^2\tilde{\alpha}(\tau,\zeta)\frac{\partial\tilde{\alpha}(\tau,\zeta)}{\partial\zeta}\mathrm{d}\tau. \tag{4.12}$$

Lemma 4.1. *If $\hat{r}^b_+$ and $\hat{\varphi}^b$ belong to $C^2[0,\xi']$, then there exist constants d, n, and δ, such that the map $\mathcal{L}$ admits a fixed point in $\Sigma(\delta)$. The constant δ depends only on m, ϑ, and the first order norms of $\hat{r}^p_-$ on $[\zeta_1,\zeta_2]$ and $H(\varphi,\omega)$ on $\big[\hat{\varphi}^b(0),\hat{\varphi}^b(\delta_0)\big]\times[\alpha^b(0,0)-\vartheta/4,\alpha^b(0,0)+\vartheta/4]$, where*

$$H(\varphi,\omega):=h\big(c(\omega,\varphi),\omega\big),\quad m:=\max\big\{\|(\hat{r}^b_+)'\|_{C^0([0,\xi'])},\|(\hat{\varphi}^b)'\|_{C^0([0,\xi'])}\big\}. \tag{4.13}$$

Proof. In order to reduce the number of constants, we will use k to denote various constants that depend only on d, ϑ, and the first order norms of $\hat{r}^p_-$ on $[\zeta_1,\zeta_2]$ and $H(\varphi,\omega)$ on $\big[\hat{\varphi}^b(0),\hat{\varphi}^b(\delta_0)\big]\times[\alpha^b(0,0)-\vartheta/4,$ $\alpha^b(0,0)+\vartheta/4]$. The proof of this lemma proceeds in three steps.

Step 1. Take any $(\xi,\zeta)\in\Lambda(\delta)$, then by integrating the first equation of system (4.11) alone ξ axis from $(0,\zeta)$ to (ξ,ζ), we have

$$\hat{r}_-(\xi,\zeta)=\hat{r}^p_-(\zeta)+\int_0^\xi\left(\frac{h}{\cos\tilde{\alpha}}\right)(\tau,\zeta)\mathrm{d}\tau. \tag{4.14}$$

Hence, we have

$$\big|\hat{r}_-(\xi,\zeta)-\hat{r}^p_-(\zeta_1)\big|\le\big|\hat{r}^p_-(\zeta)-\hat{r}^p_-(\zeta_1)\big|+\left|\int_0^\xi\left(\frac{h}{\cos\tilde{\alpha}}\right)(\tau,\zeta)\mathrm{d}\tau\right|\le k\delta. \tag{4.15}$$

Let $\tilde{\xi}=\tilde{\xi}_-(\tilde{\zeta};\xi,\zeta)$ be the characteristic curve

$$\begin{cases}\dfrac{\partial\tilde{\xi}_-(\tilde{\zeta};\xi,\zeta)}{\partial\tilde{\zeta}}=-\left(\dfrac{a\cos\tilde{\alpha}\cos\tilde{\beta}}{\sin 2\tilde{\omega}}\right)\big(\tilde{\xi}_-(\tilde{\zeta};\xi,\zeta),\tilde{\zeta}\big), & \zeta_1\le\tilde{\zeta}\le\zeta;\\ \tilde{\xi}_-(\zeta;\xi,\zeta)=\xi.\end{cases} \tag{4.16}$$

By computation, we have

$$\left|\frac{a\cos\tilde{\alpha}\cos\tilde{\beta}}{\sin 2\tilde{\omega}}\right|\le k\delta. \tag{4.17}$$

So, when δ is sufficiently small, e.g. $\delta\le 1/k$ the characteristic curve $\tilde{\xi}=\tilde{\xi}_-(\tilde{\zeta};\xi,\zeta)$ remains in $\Lambda(\delta)$ for every $(\xi,\zeta)\in\Lambda(\delta)$. Integrating the second equation of (4.11) along the characteristic line $\tilde{\xi}=\tilde{\xi}_-(\tilde{\zeta};\xi,\zeta)$ from $\big(\tilde{\xi}_-(\zeta_1;\xi,\zeta),\zeta_1\big)$ to (ξ,ζ), we have

$$\hat{r}_+(\xi,\zeta)=\hat{r}^b_+\big(\tilde{\xi}_-(\zeta_1;\xi,\zeta)\big)-\int_{\zeta_1}^\zeta\left(\frac{ah\cos\tilde{\alpha}}{\sin 2\tilde{\omega}}\right)\big(\tilde{\xi}_-(\chi;\xi,\zeta),\chi\big)\mathrm{d}\chi. \tag{4.18}$$

Hence,

$$\begin{aligned}\big|\hat{r}_+(\xi,\zeta)-\hat{r}^b_+(0)\big|&\le\big|\hat{r}^b_+\big(\tilde{\xi}_-(\zeta_1;\xi,\zeta)\big)-\hat{r}^b_+(0)\big|+\left|\int_{\zeta_1}^\zeta\left(\frac{ah\cos\tilde{\alpha}}{\sin 2\tilde{\omega}}\right)\big(\tilde{\xi}_-(\chi;\xi,\zeta),\chi\big)\mathrm{d}\chi\right|\\&\le k\delta.\end{aligned} \tag{4.19}$$

Let $\tilde{\xi} = \tilde{\xi}_\perp(\tilde{\zeta};\xi,\zeta)$ be the characteristic curve

$$\begin{cases} \dfrac{\partial \tilde{\xi}_\perp(\tilde{\zeta};\xi,\zeta)}{\partial \tilde{\zeta}} = \left(\dfrac{a\sin\tilde{\theta}}{\sin\tilde{\theta}\tan\tilde{\alpha} - \cos\tilde{\theta}}\right)\left(\tilde{\xi}_\perp(\tilde{\zeta};\xi,\zeta),\tilde{\zeta}\right), & \zeta_1 \le \tilde{\zeta} \le \zeta; \\ \tilde{\xi}_\perp(\zeta;\xi,\zeta) = \xi. \end{cases} \tag{4.20}$$

By computation, we have

$$\left|\frac{a\sin\tilde{\theta}}{\sin\tilde{\theta}\tan\tilde{\alpha} - \cos\tilde{\theta}}\right| = \left|\frac{a\sin\tilde{\theta}\cos\tilde{\alpha}}{\cos(\tilde{\theta}+\tilde{\alpha})}\right| \le k\delta, \tag{4.21}$$

since $\frac{3}{2}\alpha^b(0,0) - \frac{\pi}{2} < \tilde{\theta} + \tilde{\alpha} < \frac{1}{2}\alpha^b(0,0) + \frac{\pi}{4}$ by (4.8). So, when δ is sufficiently small, e.g. $\delta \le 1/k$, the characteristic curve $\tilde{\xi} = \tilde{\xi}_\perp(\tilde{\zeta};\xi,\zeta)$ remains in $\Lambda(\delta)$ for every $(\xi,\zeta) \in \Lambda(\delta)$. Integrating the third equation of (4.11) along the characteristic line $\tilde{\xi} = \tilde{\xi}_\perp(\tilde{\zeta};\xi,\zeta)$ from $\left(\tilde{\xi}_\perp(\zeta_1;\xi,\zeta),\zeta_1\right)$ to (ξ,ζ), we have

$$\hat{\varphi}^b(0) \le \hat{\varphi}(\xi,\zeta) = \hat{\varphi}^b\left(\tilde{\xi}_\perp(\zeta_1;\xi,\zeta)\right) \le \hat{\varphi}^b(\delta_0). \tag{4.22}$$

Therefore, by (4.15), (4.19), and (4.22) we know that when δ is sufficiently small, we have $(\hat{r}_+,\hat{r}_-)(\xi,\zeta) \in R(\delta_0)$ and $\hat{\varphi}(\xi,\zeta) \in \left[\hat{\varphi}^b(0),\hat{\varphi}^b(\delta_0)\right]$ as $(\xi,\zeta) \in \Lambda(\delta)$.

Step 2. From (4.14), we have

$$\frac{\partial \hat{r}_-(\xi,\zeta)}{\partial \zeta} = \frac{\mathrm{d}\hat{r}_-^p(\zeta)}{\mathrm{d}\zeta} + \int_0^\xi \frac{\partial}{\partial \zeta}\left(\frac{h}{\cos\tilde{\alpha}}\right)(\tau,\zeta)\mathrm{d}\tau. \tag{4.23}$$

By (2.25) and the first equation of (4.11) we have

$$\left|\frac{\partial \hat{r}_-(\xi,\zeta)}{\partial \xi}\right| \le \left|\frac{h(\tilde{c},\tilde{\omega})}{\cos\tilde{\alpha}}\right| < \frac{2(\gamma+1)}{(\gamma-1)c^b(0,0)\cos\left(\alpha^b(0,0)+\vartheta/4\right)}. \tag{4.24}$$

Thus, by (4.24) we have

$$\|D\hat{r}_-\|_{0;\Lambda(\delta)} \le 2 + \frac{2(\gamma+1)}{(\gamma-1)c^b(0,0)\cos\left(\alpha^b(0,0)+\vartheta/4\right)} + k\delta. \tag{4.25}$$

For the derivatives of $\hat{r}_+$, we note that

$$\frac{\partial \tilde{\xi}_-(\tilde{\zeta};\xi,\zeta)}{\partial \xi} = \exp\left(\int_{\tilde{\zeta}}^{\zeta} \frac{\partial}{\partial \tilde{\xi}_-}\left(\frac{a\cos\tilde{\alpha}\cos\tilde{\beta}}{\sin 2\tilde{\omega}}\right)\left(\tilde{\xi}_-(\chi;\xi,\zeta),\chi\right)\mathrm{d}\chi\right). \tag{4.26}$$

From (4.18), we have

$$\frac{\partial \hat{r}_+(\xi,\zeta)}{\partial \xi} = \frac{\mathrm{d}\hat{r}_+^b\left(\tilde{\xi}_-(\zeta_1;\xi,\zeta)\right)}{\mathrm{d}\tilde{\xi}_-} \cdot \frac{\partial \tilde{\xi}_-(\zeta_1;\xi,\zeta)}{\partial \xi} - \int_{\zeta_1}^{\zeta} \frac{\partial}{\partial \tilde{\xi}_-}\left(\frac{ah\cos\tilde{\alpha}}{\sin 2\tilde{\omega}}\right) \cdot \frac{\partial \tilde{\xi}_-(\chi,\xi,\zeta)}{\partial \xi}\mathrm{d}\chi. \tag{4.27}$$

From the second equation of (4.11) we have

$$\frac{\partial \hat{r}_+}{\partial \zeta} = \left(a\cos\tilde{\beta}\frac{\partial \hat{r}_+}{\partial \xi} - ah\right)\frac{\cos\tilde{\alpha}}{\sin 2\tilde{\omega}}. \tag{4.28}$$

Combining (4.27) and (4.28), we have

$$\|D\hat{r}_+\|_{0;\Lambda(\delta)} \le (1+m)(1+k\delta)^2. \tag{4.29}$$

For the derivatives of $\hat{\varphi}$, we note that

$$\frac{\partial\tilde{\xi}_\perp(\tilde{\zeta};\xi,\zeta)}{\partial\xi} = \exp\left(-\int_{\tilde{\zeta}}^{\zeta} \frac{\partial}{\partial\tilde{\xi}_\perp}\Big(\frac{a\sin\tilde{\theta}}{\sin\tilde{\theta}\tan\tilde{\alpha}-\cos\tilde{\theta}}\Big)(\tilde{\xi}_\perp(\chi;\xi,\zeta),\chi)\mathrm{d}\chi\right). \tag{4.30}$$

Then, by (4.22) we have

$$\frac{\partial\hat{\varphi}(\xi,\zeta)}{\partial\xi} = \frac{\mathrm{d}\hat{\varphi}^b\big(\tilde{\xi}_\perp(\zeta_1;\xi,\zeta)\big)}{\mathrm{d}\tilde{\xi}_\perp}\cdot\frac{\partial\tilde{\xi}_\perp(\zeta_1;\xi,\zeta)}{\partial\xi}. \tag{4.31}$$

From the third equation of (4.11) we have

$$\frac{\partial\hat{\varphi}}{\partial\zeta} = \frac{a\sin\tilde{\theta}}{\cos\tilde{\theta}-\sin\tilde{\theta}\tan\tilde{\alpha}}\cdot\frac{\partial\hat{\varphi}}{\partial\xi}. \tag{4.32}$$

Combining (4.31) and (4.32), we have

$$\|D\hat{\varphi}\|_{0;\Lambda(\delta)} \le (1+m)(1+k\delta)^2. \tag{4.33}$$

Thus, by letting $d = 2\Big(2+\frac{2(\gamma+1)}{(\gamma-1)c^b(0,0)\cos(\alpha^b(0,0)+\vartheta/4)}+m\Big)$ and δ sufficiently small, e.g. $\delta\le\frac{1}{4k}$, we have

$$\big\|(D\hat{r}_+, D\hat{r}_-, D\hat{\varphi})\big\|_{0;\Lambda(\delta)} \le d. \tag{4.34}$$

Step 3. For the second order derivatives of $\hat{r}_\pm$, we let

$$\hat{\partial}_+ = \cos\tilde{\alpha}\frac{\partial}{\partial\xi}, \quad \hat{\partial}_- = a\cos\tilde{\beta}\frac{\partial}{\partial\xi} - \frac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\frac{\partial}{\partial\zeta}. \tag{4.35}$$

Then, by the first equation of (4.11) we have

$$\hat{\partial}_+\hat{\partial}_+\hat{r}_- = \frac{\partial h(\tilde{c},\tilde{\omega})}{\partial\tilde{c}}\hat{\partial}_+\tilde{c} + \frac{\partial h(\tilde{c},\tilde{\omega})}{\partial\tilde{\omega}}\hat{\partial}_+\tilde{\omega}, \quad \hat{\partial}_-\hat{\partial}_+\hat{r}_- = \frac{\partial h(\tilde{c},\tilde{\omega})}{\partial\tilde{c}}\hat{\partial}_-\tilde{c} + \frac{\partial h(\tilde{c},\tilde{\omega})}{\partial\tilde{\omega}}\hat{\partial}_-\tilde{\omega}. \tag{4.36}$$

By direct computation, we get

$$\begin{aligned}\hat{\partial}_+\hat{\partial}_-\hat{r}_- &= \hat{\partial}_-\hat{\partial}_+\hat{r}_- + \cos\tilde{\alpha}\frac{\partial(a\cos\tilde{\beta})}{\partial\xi}\cdot\frac{\partial\hat{r}_-}{\partial\xi} - \cos\tilde{\alpha}\frac{\partial}{\partial\xi}\left(\frac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\right)\cdot\frac{\partial\hat{r}_-}{\partial\zeta} \\ &\quad - a\cos\tilde{\beta}\frac{\partial(\cos\tilde{\alpha})}{\partial\xi}\cdot\frac{\partial\hat{r}_-}{\partial\xi} + \frac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\cdot\frac{\partial(\cos\tilde{\alpha})}{\partial\zeta}\cdot\frac{\partial\hat{r}_-}{\partial\xi} \\ &= W_1\hat{\partial}_-\hat{r}_- + W_2 + W_3\hat{\partial}_+\hat{r}_-,\end{aligned} \tag{4.37}$$

where

$$W_1 = \frac{\cos^2\tilde{\alpha}}{\sin 2\tilde{\omega}}\cdot\frac{\partial}{\partial\xi}\left(\frac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\right), \quad W_2 = \frac{\partial h(\tilde{c},\tilde{\omega})}{\partial\tilde{c}}\hat{\partial}_-\tilde{c} + \frac{\partial h(\tilde{c},\tilde{\omega})}{\partial\tilde{\omega}}\hat{\partial}_-\tilde{\omega},$$

$$W_3 = \frac{\partial(a\cos\tilde{\beta})}{\partial\xi} - \frac{a\cos\tilde{\beta}}{\cos\tilde{\alpha}}\frac{\partial(\cos\tilde{\alpha})}{\partial\xi} + \frac{\sin 2\tilde{\omega}}{\cos^2\tilde{\alpha}}\frac{\partial(\cos\tilde{\alpha})}{\partial\zeta} - \frac{a\cos\tilde{\beta}\cos\tilde{\alpha}}{\sin 2\tilde{\omega}}\frac{\partial}{\partial\xi}\left(\frac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\right).$$

By computation we also have

$$\begin{aligned}\hat{\partial}_+\hat{\partial}_-\hat{\partial}_-\hat{r}_- &= \hat{\partial}_-\hat{\partial}_+\hat{\partial}_-\hat{r}_- + W_1\hat{\partial}_-\hat{\partial}_-\hat{r}_- + W_3\hat{\partial}_+\hat{\partial}_-\hat{r}_- \\ &= 2W_1\hat{\partial}_-\hat{\partial}_-\hat{r}_- + \hat{\partial}_-W_1\cdot\hat{\partial}_-\hat{r}_- + \hat{\partial}_-W_2 + \hat{\partial}_-W_3\cdot\hat{\partial}_+\hat{r}_- + W_2W_3 \\ &\quad + W_3(W_1\hat{\partial}_-\hat{r}_- + W_2 + W_3\hat{\partial}_+\hat{r}_-).\end{aligned} \tag{4.38}$$

Integrating (4.38) along $\zeta = \text{Const.}$ from $(0,\zeta)$ to (ξ,ζ) we have

$$\|\hat{\partial}_-\hat{\partial}_-r_-\|_{0;\Lambda(\delta)} \le k\delta(1+n)(1+k\delta) + (1+k\delta)\max_{\zeta\in[\zeta_1,\zeta_2]}\left|(\hat{r}_-^p)''(\zeta)\right|. \tag{4.39}$$

Consequently, by (4.36)–(4.37) we have

$$\|D^2r_-\|_{0;\Lambda(\delta)} \le k + k\delta(1+n)(1+k\delta) + k(1+k\delta)\max_{\xi\in[0,\delta]}\left|(\hat{r}_-^p)''(\zeta)\right|. \tag{4.40}$$

Similarly, we have

$$\|D^2\hat{r}_+\|_{0;\Lambda(\delta)} \le k + k\delta(1+n)(1+k\delta) + (1+k\delta)\max_{\xi\in[0,\xi']}\left|(\hat{r}_+^b)''(\xi)\right| \tag{4.41}$$

and

$$\|D^2\hat{\varphi}\|_{0;\Lambda(\delta)} \le k + k\delta(1+n)(1+k\delta) + (1+k\delta)\max_{\xi\in[0,\xi']}\left|(\hat{\varphi}^b)''(\xi)\right|. \tag{4.42}$$

So, by letting

$$n = 2\Big(k\max_{\xi\in[0,\xi']}\left|(\hat{r}_+^b)''(\xi)\right| + k\max_{\xi\in[0,\xi']}\left|(\hat{\varphi}^b)''(\xi)\right| + k\max_{\zeta\in[\zeta_1,\zeta_2]}\left|(\hat{r}_-^p)''(\zeta)\right| + k\Big) \tag{4.43}$$

and δ sufficiently small, δ depends only on k, we have $\|(D^2\hat{r}_-, D^2\hat{r}_+, D^2\hat{\varphi})\|_{0;\Lambda(\delta)} \le n$. Thus, we have $\mathcal{L}$ maps $\Sigma(\delta)$ into $\Sigma(\delta)$.

Since $\Sigma(\delta)$ is a compact convex set in $C^1(\Lambda(\delta))$, in order to use Schauder fixed point theorem it only remains to show that the operator $\mathcal{L}$ is continuous. In order to show the continuity of $\mathcal{L}$ we let $(\tilde{r}_+^j, \tilde{r}_-^j, \tilde{\varphi}^j) \in \Sigma(\delta)$, $j = 1, 2, \cdots$, converge to $(\tilde{r}_+, \tilde{r}_-, \tilde{\varphi}) \in \Sigma(\delta)$ in $C^1(\Lambda(\delta))$ as $j \to \infty$. Then, since the sequence $(\hat{r}_+^j, \hat{r}_-^j, \hat{\varphi}^j) = \mathcal{L}(\tilde{r}_+^j, \tilde{r}_-^j, \tilde{\varphi}^j)$ is precompact in $C^1(\Lambda(\delta))$, every subsequence in turn has a convergent subsequence. Let $(\hat{r}_+^{j_k}, \hat{r}_-^{j_k}, \hat{\varphi}^{j_k})$ be such a convergent subsequence with limit $(\hat{r}_+, \hat{r}_-, \hat{\varphi}) \in \Sigma(\delta)$. Then, since

$$\begin{cases} \cos\tilde{\alpha}\dfrac{\partial\hat{r}_-}{\partial\xi} - h(\tilde{c},\ \tilde{\omega}) = \lim\limits_{k\to\infty}\left(\cos\tilde{\alpha}^{j_k}\dfrac{\partial\hat{r}_-^{j_k}}{\partial\xi} - h(\tilde{c}^{j_k},\ \tilde{\omega}^{j_k})\right) = 0, \\ a\cos\tilde{\beta}\dfrac{\partial\hat{r}_+}{\partial\xi} - \dfrac{\sin 2\tilde{\omega}}{\cos\tilde{\alpha}}\dfrac{\partial\hat{r}_+}{\partial\zeta} - ah(\tilde{c},\ \tilde{\omega}) \\ \qquad = \lim\limits_{k\to\infty}\left(a^{j_k}\cos\tilde{\beta}^{j_k}\dfrac{\partial\hat{r}_+^{j_k}}{\partial\xi} - \dfrac{\sin 2\tilde{\omega}^{j_k}}{\cos\tilde{\alpha}^{j_k}}\dfrac{\partial\hat{r}_+^{j_k}}{\partial\zeta} - a^{j_k}h(\tilde{c}^{j_k},\ \tilde{\omega}^{j_k})\right) = 0, \\ -a\sin\tilde{\theta}\dfrac{\partial\hat{\varphi}}{\partial\xi} + (\cos\tilde{\theta} - \sin\tilde{\theta}\tan\tilde{\alpha})\dfrac{\partial\hat{\varphi}}{\partial\zeta} \\ \qquad = \lim\limits_{k\to\infty}\left(-a^{j_k}\sin\tilde{\theta}^{j_k}\dfrac{\partial\hat{\varphi}^{j_k}}{\partial\xi} + (\cos\tilde{\theta}^{j_k} - \sin\tilde{\theta}^{j_k}\tan\tilde{\alpha}^{j_k})\dfrac{\partial\hat{\varphi}^{j_k}}{\partial\zeta}\right) = 0, \end{cases} \tag{4.44}$$

where

$$a^{j_k} = a^{j_k}(\xi,\zeta) = \int_0^{\xi} \sec^2 \tilde{\alpha}^{j_k}(\tau,\zeta)\frac{\partial \tilde{\alpha}^{j_k}(\tau,\zeta)}{\partial \zeta}\mathrm{d}\tau, \tag{4.45}$$

we must have $(\hat{r}_+, \hat{r}_-, \hat{\varphi}) = \mathcal{L}(\tilde{r}_+, \tilde{r}_-, \tilde{\varphi})$. Then the theorem is complete. □

In what follows, we are going to show that if $\hat{r}^b_+$ and $\hat{\varphi}^b$ belong to $C^1[0,\xi']$, problem (3.4), (4.2)–(4.3) still admits a local classical solution. We need the concept of modulus of continuity introduced by Hartman and Wintner [4] (see also Zheng [25]). For any function $f(t_1, t_2, \cdot, \cdot, \cdot, t_m)$ on a domain $S \subset R^m$, let

$$\begin{aligned} s(h; f) &:= \sup |f(t_1, t_2, \cdots, t_m) - f(t'_1, t'_2, \cdots, t'_m)|, \\ &\qquad (t_1, t_2, \cdots, t_m), (t'_1, t'_2, \cdots, t'_m) \in S, |t_k - t'_k| \le h, k = 1, 2, \cdots, m, \end{aligned} \tag{4.46}$$

which is referred to as the modulus of continuity of f on S. It is apparent that f is uniformly continuous if and only if $s(h; f) \to 0 (h \to 0)$; a family of $\{f\}$ is equicontinuous if and only if there is an $s(h)$ with the property $s(h) \to 0 (h \to 0)$ such that $s(h; f) \le s(h)$ for every f in the family. Other properties are:

1. if f is C^1 continuous, then $s(h; f) \le \|f\|_1 h$;
2. $s(h; f \pm g) \le s(h; f) + s(h; g)$;
3. $s(h; fg) \le \|f\|_0 s(h; g) + \|g\|_0 s(h; f)$;
4. $s(h; f/g) \le \frac{\|f\|_0}{a^2} s(h; g) + \frac{1}{a} s(h; f)$ for a scalar function g with $|g| > a > 0$;
5. $s(h; f(g)) \le s\big(s(h; g); f\big) \le \|f\|_1 s(h; g)$;
6. $s(h; F) \le \delta s(h; f) + \|f\|_0[s(h; \varphi) + s(h; \psi)]$ for $F(x, y) = \int_{\psi(x,y)}^{\varphi(x,y)} f(t, x, y)dt$ $(0 \le \phi, \varphi \le \delta)$;
7. $s(dh; f) \le ([d] + 1)s(h; f)$, $d > 0$ is a constant, $[d]$ represents the integer part of d.

If $(\hat{r}^b_+, \hat{\varphi}^b) \in C^1[0,\xi']$, then there exists a sequence $(\hat{r}^b_{+,l}, \hat{\varphi}^b_l)$, $l = 1, 2, \ldots$, in $C^2[0,\xi']$, such that this sequence converges to $(\hat{r}^b_+, \hat{\varphi}^b)$ in $C^1[0,\xi']$. Then, by Lemma 4.1 we know that there exists a uniform $\delta > 0$ with respect to l, such that for any l, Eqs. (3.4) with the boundary condition (4.3) and boundary condition

$$\hat{\varphi} = \hat{\varphi}^b_l(\xi) \quad \text{and} \quad \hat{r}_+ = \hat{r}^b_{+,l}(\xi) \quad \text{on} \quad \{(\xi,\zeta_1) \mid 0 \le \xi \le \xi'\} \tag{4.47}$$

admits a solution $(\hat{r}^l_+, \hat{r}^l_-, \hat{\varphi}^l)$ on $\Lambda(\delta)$. We have the following lemma.

Lemma 4.2. *There exists a $\delta > 0$, δ dependents only m, ϑ, and the first order norms of $\hat{r}^p_-$ on $[\zeta_1, \zeta_2]$ and $H(\varphi,\omega)$ on $\big[\hat{\varphi}^b(0), \hat{\varphi}^b(\delta_0)\big] \times [\alpha^b(0,0) - \vartheta/4, \alpha^b(0,0) + \vartheta/4]$, such that the sequences $\{\partial_\xi \hat{r}^l_+\}$, $\{\partial_\xi \hat{r}^l_-\}$, $\{\partial_\zeta \hat{r}^l_+\}$, $\{\partial_\zeta \hat{r}^l_-\}$, $\{\partial_\xi \hat{\varphi}^l\}$, and $\{\partial_\zeta \hat{\varphi}^l\}$ are equicontinuous on $\Lambda(\delta)$.*

Proof. Let $\hat{\xi} = \hat{\xi}^l_-(\hat{\zeta}; \xi, \zeta)$ be the characteristic curve

$$\begin{cases} \dfrac{\partial \hat{\xi}^l_-(\hat{\zeta}; \xi, \zeta)}{\partial \hat{\zeta}} = \hat{\lambda}^l_-(\hat{\xi}^l_-(\hat{\zeta}; \xi, \zeta), \hat{\zeta}), & \zeta_1 \le \hat{\zeta} \le \zeta; \\ \hat{\xi}^l_-(\zeta; \xi, \zeta) = \xi, \end{cases} \tag{4.48}$$

where

$$\hat{\lambda}^l_-(\hat{\xi}^l_-(\hat{\zeta}; \xi, \zeta), \hat{\zeta}) = -\left(\frac{g^l_\zeta \cos\hat{\beta}^l \cos\hat{\alpha}^l}{\sin 2\hat{\omega}^l}\right)(\hat{\xi}^l_-(\hat{\zeta}; \xi, \zeta), \hat{\zeta}), \tag{4.49}$$

$$g^l_\zeta(\hat{\xi}^l_-(\hat{\zeta}; \xi, \zeta), \hat{\zeta}) = \int_0^{\hat{\xi}^l_-(\hat{\zeta};\xi,\zeta)} \sec^2 \hat{\alpha}^l(\tau, \hat{\zeta}) \frac{\partial \hat{\alpha}^l(\tau, \hat{\zeta})}{\partial \hat{\zeta}} \mathrm{d}\tau, \tag{4.50}$$

$\hat{\alpha}^l = \alpha(\hat{r}^l_+, \hat{r}^l_-)$, and $\hat{\beta}^l = \beta(\hat{r}^l_+, \hat{r}^l_-)$. Then we have

$$\frac{\partial \hat{\xi}^l_-(\hat{\zeta};\xi,\zeta)}{\partial \xi} = \exp\left(-\int_{\hat{\zeta}}^{\zeta} \frac{\partial \hat{\lambda}^l_-}{\partial \hat{\xi}^l_-}\big(\hat{\xi}^l_-(\chi;\xi,\zeta),\chi\big)\mathrm{d}\chi\right) \tag{4.51}$$

and

$$\frac{\partial \hat{\xi}^l_-(\hat{\zeta};\xi,\zeta)}{\partial \zeta} = -\hat{\lambda}^l_-(\xi,\zeta)\cdot \exp\left(-\int_{\hat{\zeta}}^{\zeta} \frac{\partial \hat{\lambda}^l_-}{\partial \hat{\xi}^l_-}\big(\hat{\xi}^l_-(\chi;\xi,\zeta),\chi\big)\mathrm{d}\chi\right). \tag{4.52}$$

By computation, we have the derivatives of $\hat{r}^l_+$,

$$\begin{aligned}\frac{\partial \hat{r}^l_+(\xi,\zeta)}{\partial \xi} &= \frac{\mathrm{d}\hat{r}^b_{+,l}\big(\hat{\xi}^l_-(\zeta_1;\xi,\zeta)\big)}{\mathrm{d}\hat{\xi}^l_-}\cdot\frac{\partial \hat{\xi}^l_-(\zeta_1;\xi,\zeta)}{\partial \xi}\\ &\quad -\int_{\zeta_1}^{\zeta}\frac{\partial}{\partial \hat{\xi}^l_-}\left(\Big(\frac{g^l_\zeta h^l \cos\hat{\alpha}^l}{\sin 2\hat{\omega}}\Big)(\hat{\xi}^l_-(\chi,\xi,\zeta),\chi)\right)\cdot\frac{\partial \hat{\xi}^l_-(\chi,\xi,\zeta)}{\partial \xi}\mathrm{d}\chi\end{aligned} \tag{4.53}$$

and

$$\frac{\partial \hat{r}^l_+(\xi,\zeta)}{\partial \zeta} = \Big(g^l_\zeta(\xi,\zeta)\cos\hat{\beta}^l(\xi,\zeta)\frac{\partial \hat{r}^l_+(\xi,\zeta)}{\partial \xi} - g^l_\zeta(\xi,\zeta)h^l(\xi,\zeta)\Big)\cdot\frac{\cos\hat{\alpha}^l(\xi,\zeta)}{\sin 2\hat{\omega}^l(\xi,\zeta)}, \tag{4.54}$$

where $h^l = h\big(c(\hat{\omega}^l,\hat{\varphi}^l),\hat{\omega}^l\big)$.

For the derivatives of $\hat{r}^l_-$, we have

$$\frac{\partial \hat{r}^l_-(\xi,\zeta)}{\partial \zeta} = \frac{\mathrm{d}\hat{r}^p_-(\zeta)}{\mathrm{d}\zeta} + \int_0^{\xi}\frac{\partial}{\partial \zeta}\Big(\frac{h^l}{\cos\hat{\alpha}^l}\Big)(\tau,\zeta)\mathrm{d}\tau, \quad \frac{\partial \hat{r}^l_-}{\partial \xi} = \frac{h\big(c(\hat{\omega}^l,\hat{\varphi}^l),\hat{\omega}^l\big)}{\cos\hat{\alpha}^l}. \tag{4.55}$$

Let $\hat{\xi} = \hat{\xi}^l_-(\hat{\zeta};\xi,\zeta)$ be the characteristic curve

$$\begin{cases}\dfrac{\partial \hat{\xi}^l_\perp(\hat{\zeta};\xi,\zeta)}{\partial \hat{\zeta}} = \hat{\lambda}^l_\perp\big(\hat{\xi}^l_-(\hat{\zeta};\xi,\zeta),\hat{\zeta}\big), & \zeta_1 \le \tilde{\zeta} \le \zeta;\\ \hat{\xi}^l_\perp(\zeta;\xi,\zeta) = \xi, \end{cases} \tag{4.56}$$

where

$$\hat{\lambda}^l_\perp\big(\hat{\xi}^l_-(\hat{\zeta};\xi,\zeta),\hat{\zeta}\big) := \left(\frac{g^l_\zeta \sin\hat{\theta}^l}{\sin\hat{\theta}^l\tan\hat{\theta}^l - \cos\hat{\theta}^l}\right)\big(\hat{\xi}^l_\perp(\hat{\zeta};\xi,\zeta),\hat{\zeta}\big). \tag{4.57}$$

Then we have

$$\frac{\partial \hat{\xi}^l_\perp(\hat{\zeta};\xi,\zeta)}{\partial \xi} = \exp\left(-\int_{\hat{\zeta}}^{\zeta} \frac{\partial \hat{\lambda}^l_\perp}{\partial \hat{\xi}^l_\perp}\big(\hat{\xi}^l_\perp(\chi;\xi,\zeta),\chi\big)\mathrm{d}\chi\right) \tag{4.58}$$

and

$$\frac{\partial \hat{\xi}^l_\perp(\hat{\zeta};\xi,\zeta)}{\partial \zeta} = -\hat{\lambda}^l_\perp(\xi,\zeta)\cdot \exp\left(-\int_{\hat{\zeta}}^{\zeta} \frac{\partial \hat{\lambda}^l_\perp}{\partial \hat{\xi}^l_\perp}\big(\hat{\xi}^l_\perp(\chi;\xi,\zeta),\chi\big)\mathrm{d}\chi\right). \tag{4.59}$$

By computation we have the derivatives of $\hat{\varphi}^l$,

$$\frac{\partial \hat{\varphi}^l(\xi,\zeta)}{\partial \xi} = \frac{\mathrm{d}\hat{\varphi}^b\big(\hat{\xi}^l_\perp(\zeta_1;\xi,\zeta)\big)}{\mathrm{d}\tilde{\xi}^l_\perp}\cdot\frac{\partial \hat{\xi}^l_\perp(\zeta_1;\xi,\zeta)}{\partial \xi}, \quad \frac{\partial \hat{\varphi}^l}{\partial \zeta} = \frac{g^l_\zeta \sin\hat{\theta}^l}{\cos\hat{\theta}^l - \sin\hat{\theta}^l\tan\hat{\alpha}^l}\cdot\frac{\partial \hat{\varphi}^l}{\partial \xi}. \tag{4.60}$$

Thus by properties 1–7 of the modulus of continuity and that $\{(\hat{r}^b_{+,l})'\}$ and $\{(\hat{\varphi}^b_l)'\}$ are equicontinuous, we have that there exists a positive continuous function $s(h)$, $s(h)\to 0$ $(h\to 0)$, such that

$$\begin{aligned}
&s\Big(h;\frac{\partial \hat{r}^l_+}{\partial \xi}\Big)\le ks(h)+k\delta F(h), \quad s\Big(h;\frac{\partial \hat{r}^l_+}{\partial \zeta}\Big)\le ks(h)+k\delta F(h),\\
&s\Big(h;\frac{\partial \hat{r}^l_-}{\partial \xi}\Big)\le ks(h)+k\delta F(h), \quad s\Big(h;\frac{\partial \hat{r}^l_-}{\partial \zeta}\Big)\le ks(h)+k\delta F(h),\\
&s\Big(h;\frac{\partial \hat{\varphi}^l}{\partial \xi}\Big)\le ks(h)+k\delta F(h), \quad s\Big(h;\frac{\partial \hat{\varphi}^l}{\partial \zeta}\Big)\le ks(h)+k\delta F(h),
\end{aligned} \tag{4.61}$$

where

$$F(h) = s\Big(h;\frac{\partial \hat{r}^l_+}{\partial \xi}\Big)+s\Big(h;\frac{\partial \hat{r}^l_+}{\partial \zeta}\Big)+s\Big(h;\frac{\partial \hat{r}^l_-}{\partial \xi}\Big)+s\Big(h;\frac{\partial \hat{r}^l_-}{\partial \zeta}\Big)+s\Big(h;\frac{\partial \hat{\varphi}^l}{\partial \xi}\Big)+s\Big(h;\frac{\partial \hat{\varphi}^l}{\partial \zeta}\Big), \tag{4.62}$$

k is a constant which depends only on d, ϑ, and the first order norms of $\hat{r}^p_-$ on $[\zeta_1,\zeta_2]$ and $H(\varphi,\omega)$ on $\big[\hat{\varphi}^b(0),\hat{\varphi}^b(\delta_0)\big]\times[\alpha^b(0,0)-\vartheta/4,\alpha^b(0,0)+\vartheta/4]$. Thus, let δ be small, e.g. $\delta<\frac{1}{12k}$, we have

$$s\Big(h;\frac{\partial \hat{r}^l_+}{\partial \xi}\Big)+s\Big(h;\frac{\partial \hat{r}^l_+}{\partial \zeta}\Big)+s\Big(h;\frac{\partial \hat{r}^l_-}{\partial \xi}\Big)+s\Big(h;\frac{\partial \hat{r}^l_-}{\partial \zeta}\Big)+s\Big(h;\frac{\partial \hat{\varphi}^l}{\partial \xi}\Big)+s\Big(h;\frac{\partial \hat{\varphi}^l}{\partial \zeta}\Big)\le 12ks(h). \tag{4.63}$$

Then by properties of modulus of continuous we have this lemma. □

From Lemma 4.2 we immediately have that the sequence $(\hat{r}^l_+,\hat{r}^l_-,\hat{\varphi}^l)$ has a convergent subsequence $(\hat{r}^{l_j}_+,\hat{r}^{l_j}_-,\hat{\varphi}^{l_j})$ with a limit $(\hat{r}_+,\hat{r}_-,\hat{\varphi})$ in $C^1(\Lambda(\delta))$, and $(\hat{r}_+,\ \hat{r}_-,\hat{\varphi})$ is a solution of problem (3.4), (4.2)–(4.3) on $\Lambda(\delta)$.

Since $\sec^2\hat{\alpha}(0,\zeta)\frac{\partial\hat{\alpha}}{\partial\zeta}(0,\zeta)=1$, there exists a sufficiently small $\delta_\xi>0$, such that $\frac{\partial g(\xi,\zeta)}{\partial \zeta}>0$ in

$$\Lambda(\delta,\delta_\xi) := \left\{(\xi,\zeta)\mid \zeta\ge\zeta_1,\ \ \frac{\xi}{\delta_\xi}+\frac{\zeta-\zeta_1}{\delta}\le 1\right\}. \tag{4.64}$$

Thus, we have the following theorem.

Theorem 4.3 *(Local C^1 solution to the centered wave bubble problem). Let $\widetilde{OB}$ be a C_+ characteristic curve passing through the origin with the slope $\zeta_1+\delta$ at the origin, A be a point on $\widetilde{OC}$ with $\xi(A)=\delta_\xi$, $\widetilde{BA}$ be a C_- characteristic line passing through A, then for sufficiently small δ and δ_ξ, δ depends only on m and ϑ, the centered wave problem (2.18), (3.7), and (3.8) admits a solution on a closed domain $\Omega^c(\delta)$ bounded by $\widetilde{OA}$, $\widetilde{OB}$, and $\widetilde{BA}$, and this solution belongs to $C^1\big(\Omega^c(\delta)\setminus\{(0,0)\}\big)$. See Fig. 4 (left).*

4.2. Goursat problems regarded as subproblems

According to Theorem 4.3, we obtain the C_- characteristic curve $\widetilde{BA}$ and the (α, β, c) on it. In what follows, we are going to construct solution to system (2.18) with boundary conditions

$$\begin{cases} (\alpha, \beta, \varphi) = \left(\alpha_{BA},\ \beta_{BA},\ \varphi_{BA}\right) & \text{on } \widetilde{BA}; \\ (\alpha, \beta, \varphi) = \left(\alpha^b,\ \beta^b,\ \varphi^b\right) & \text{on } \widetilde{AC}, \end{cases} \tag{4.65}$$

where $\left(\alpha_{BA},\ \beta_{BA},\ \varphi_{BA}\right)$ is the value of the local solution on $\widetilde{BA}$.

By computation, we have

$$\bar{\partial}_- \zeta = -\left(\frac{\partial g}{\partial \zeta}\right)^{-1} \frac{\sin 2\hat{\omega}}{\cos \hat{\alpha}} < 0. \tag{4.66}$$

So, the direction of the characteristic curve $\widetilde{BA}$ is B points toward A, since the ζ at the point B is greater then the ζ at the point A. By computation, we have

$$\bar{\partial}_- \alpha = \cos \hat{\beta} \frac{\partial \hat{\alpha}}{\partial \xi} - \left(\frac{\partial g}{\partial \zeta}\right)^{-1} \frac{\sin 2\hat{\omega}}{\cos \hat{\alpha}} \frac{\partial \hat{\alpha}}{\partial \zeta}. \tag{4.67}$$

Thus, according to $\hat{\alpha}(0, \zeta) = \arctan \zeta$ we have

$$\bar{\partial}_- \alpha_{BA} \mid_{\widetilde{BA}} < 0, \tag{4.68}$$

as δ_ξ is sufficiently small. By (2.22) we also have

$$\bar{\partial}_- c \mid_{\widetilde{BA}} < 0. \tag{4.69}$$

4.2.1. Local solution

Since problem (2.18), (4.65) is a Goursat-type boundary value problem, we have the following theorem.

Lemma 4.4 *(Local solution to the Goursat problem). For a fixed ω_l ($\omega_l > \omega(A) = (\alpha^b(A) - \beta^b(A))/2$), when ω_l is sufficiently close to $\omega(A)$, the Goursat problem (2.18), (4.65) admits a unique C^1 solution on a domain $\Omega^G_{\omega_l}(\delta)$ (see Fig. 6 (left)) closed by $\widetilde{AC}$, $\widetilde{BA}$, and a level curve $\omega(\xi, \eta) = \omega_l$. This solution satisfies*

$$\begin{aligned} &\Omega < 0, \quad 3\tan^2 \omega > 1, \quad \bar{\partial}_+ c > 0,\ \bar{\partial}_- c < 0,\ \bar{\partial}_+ \alpha > 0, \\ &\qquad\qquad \bar{\partial}_- \alpha < 0,\ \bar{\partial}_+ \beta < 0,\ \bar{\partial}_- \beta > 0,\ \bar{\partial}_+ \omega > 0,\ \bar{\partial}_- \omega < 0. \end{aligned} \tag{4.70}$$

Proof. By Li and Yu [12] we know that the Goursat problem admits a unique local C^1 solution. From $\bar{\partial}_+ c^b > 0$, $\omega^b(0, 0) > \max\left\{\arctan\sqrt{\frac{3-\gamma}{1+\gamma}},\ \arctan\frac{1}{3}\right\}$, (2.19)–(2.22), (2.27), (2.32), (4.68), and (4.69) we have (4.70), since the direction the C_+ characteristic curve $\widetilde{AC}$ is A points toward C and the direction of the characteristic curve $\widetilde{BA}$ is B points toward A, we have Lemma 4.4. $\square$

Remark 4.5. Estimate (4.70) implies that each level curve $\omega(\xi, \eta) = \omega_l$ intersects with any characteristic curve at most one point.

The following are some illustrations about domain $\Omega^G_{\omega_l}(\delta)$:

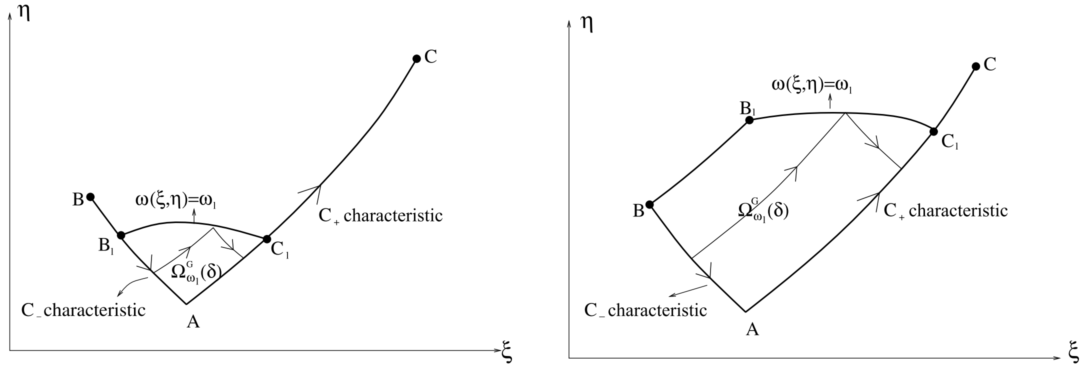

Fig. 6. Local solution of the Goursat problem on $\Omega^G_{\omega_l}(\delta)$.

- If $\omega(A) < \omega_l \leq \omega(B)$, we let C_l and B_l be respectively the points on $\widetilde{AC}$ and $\widetilde{BA}$, such that $\omega(B_l) = \omega(C_l) = \omega_l$. Then we denote by $\Omega^G_{\omega_l}(\delta)$ the domain closed by $\widetilde{AC_l}$, $\widetilde{B_lA}$, and a level curve $\omega(\xi,\eta) = \omega_l$ connecting B_l and C_l; see Fig. 6 (left).
- If $\omega(B) < \omega_l < \frac{\pi}{2}$ we let $\widetilde{BB_l}$ be the C_+ characteristic curve passing through B with $\omega(B_l) = \omega_l$. Then we denote by $\Omega^G_{\omega_l}(\delta)$ the domain closed by $\widetilde{AC_l}$, $\widetilde{BA}$, $\widetilde{BB_l}$, and a level curve $\omega(\xi,\eta) = \omega_l$ connecting B_l and C_l; see Fig. 6 (right).

4.2.2. C^0 estimate and hyperbolicity

Lemma 4.6. *Assume that the Goursat problem admits a C^1 solution on $\Omega^G_{\omega_l}(\delta)$ where $\omega(A) < \omega_l < \frac{\pi}{2}$, then*

$$(\alpha,\beta) \in \big\{(\alpha,\beta) \mid \alpha > \alpha(A),\ \beta < \beta(A),\ \alpha - \beta < 2\omega_l\big\} \quad \text{in } \Omega^G_{\omega_l}(\delta). \tag{4.71}$$

Proof. This lemma can be obtained by (4.70). □

Lemma 4.7. *Assume that the Goursat problem admits a C^1 solution on $\Omega^G_{\omega_l}(\delta)$ where $\omega(A) < \omega_l < \frac{\pi}{2}$, then*

$$c^b(0,0) \leq c < c_M := c^b(0,0)\sqrt{\frac{(\gamma-1) + 2\sin^2\omega^b(0,0)}{(\gamma-1)\sin^2\omega^b(0,0)}} \cdot e^{\frac{\pi\mu^2}{\sin^3\omega^b(0,0)}} \qquad \text{in } \Omega^G_{\omega_l}(\delta). \tag{4.72}$$

Proof. From (2.19) and (2.20), we have

$$c\bar{\partial}_+(\alpha-\beta) = 2\sin^2\omega + \frac{\sin\omega}{\mu^2}\left(\tan^2\omega\cos\omega + \frac{1}{\cos\omega} - \frac{3-\gamma}{\gamma+1}\cos\omega\right)\bar{\partial}_+ c. \tag{4.73}$$

Hence, by $1 < \gamma < 3$ and $\bar{\partial}_+ c > 0$ we have

$$\frac{\bar{\partial}_+ c}{c} < \frac{2\mu^2\bar{\partial}_+\omega}{\sin^3\omega}. \tag{4.74}$$

Let S be an arbitrary point in $\Omega^G_{\omega_l}(\delta)$, $\widetilde{OS}$ be the C_+ characteristic curve passing through S with a slope ζ ($\zeta_1 < \zeta \leq \zeta_1 + \delta$) at the origin O. Then by integrating (4.74) along $\widetilde{OS}$ from O to S and noting that $\bar{\partial}_+\omega > 0$ along $\widetilde{OS}$ and $\hat{\omega}^p(\zeta) > \omega^b(0,0)$, we have

$$c(S) < \hat{c}^p(\zeta)e^{\frac{\pi\mu^2}{\sin^3\omega^b(0,0)}} \tag{4.75}$$

in which $\hat{c}^p(\zeta) = c\big(\hat{\omega}^p(\zeta), \hat{\varphi}^p(\zeta)\big)$. From

$$\frac{\left(\hat{c}^p(\zeta)\right)^2}{2\sin^2\hat{\omega}^p(\zeta)}+\frac{\left(\hat{c}^p(\zeta)\right)^2}{\gamma-1}=\frac{\left(c^b(0,0)\right)^2}{2\sin^2\omega^b(0,0)}+\frac{\left(c^b(0,0)\right)^2}{\gamma-1},\tag{4.76}$$

we have

$$\hat{c}^p(\zeta)<c^b(0,0)\sqrt{\frac{(\gamma-1)+2\sin^2\omega^b(0,0)}{(\gamma-1)\sin^2\omega^b(0,0)}}.\tag{4.77}$$

We then have Lemma 4.7. □

4.2.3. Gradient estimate

Lemma 4.8. *Assume that the Goursat problem admits a C^1 solution on $\Omega^G_{\omega_l}(\delta)$, where $\omega(A)<\omega_l<\frac{\pi}{2}$, then for any $\varepsilon>0$, we have*

$$\left(\frac{\bar{\partial}_+c}{\sin^2\omega},\ \frac{\bar{\partial}_-c}{\sin^2\omega}\right)\in(0,\varrho_{\omega_l}+\varepsilon)\times(-\varrho_{\omega_l}-\varepsilon,0)\quad in\ \Omega^G_{\omega_l}(\delta),\tag{4.78}$$

where

$$\varrho_{\omega_l}:=\max\left\{\sup_{\widetilde{AC_l}}\frac{\bar{\partial}_+c}{\sin^2\omega},\quad\sup_{\widetilde{BA}}\frac{-\bar{\partial}_-c}{\sin^2\omega}\right\}.\tag{4.79}$$

Proof. Estimate (4.78) was first obtained by Li and Zheng [10]. In what follows, we shall give a simplified proof based on a idea in Song and Zheng [17] for the sake of completeness.

Since $\bar{\partial}_+c\,|_{\widetilde{AC}}>0$ and $\bar{\partial}_-c\,|_{\widetilde{BA}}<0$, by virtue of the characteristic decompositions (2.35) we have

$$\bar{\partial}_+c>0\quad\text{and}\quad\bar{\partial}_-c<0\quad\text{in}\ \ \Omega^G_{\omega_l}(\delta).\tag{4.80}$$

We shall use the method of proof by contradiction. For convenience, we let $\gamma_1=(0,\varrho+\varepsilon)\times\{-\varrho-\varepsilon\}$, $\gamma_2=\{\varrho+\varepsilon\}\times(-\varrho-\varepsilon,0)$, and $\gamma_3=\{(\varrho+\varepsilon,-\varrho-\varepsilon)\}$. Suppose that this proposition is not valid, then there exists a point G in $\Omega^G_{\omega_l}(\delta)$, such that $\left(\frac{\bar{\partial}_+c(G)}{\sin^2\omega(G)},\ \frac{\bar{\partial}_-c(G)}{\sin^2\omega(G)}\right)\in\bigcup_{i=1}^{3}\gamma_i$ and $\left(\frac{\bar{\partial}_+c(\xi,\eta)}{\sin^2\omega(\xi,\eta)},\ \frac{\bar{\partial}_-c(\xi,\eta)}{\sin^2\omega(\xi,\eta)}\right)\in(0,\varrho+\varepsilon)\times(-\varrho-\varepsilon,0)$ for any $(\xi,\eta)\in\Sigma_G\setminus\{G\}$. Here, the C_+ characteristic curve passing through G intersects with $\widetilde{BA}$ at a point B_G, the C_- characteristic curve passing through G intersects with $\widetilde{AC}$ at a point C_G, and Σ_G represents the closed domain bounded by $\widetilde{AB_G}$, $\widetilde{AC_G}$, $\widetilde{GB_G}$, and $\widetilde{GC_G}$; see Fig. 7 (right). (Roughly speaking, suppose that G is the "first" point to reach the boundary of $(0,\varrho_{\omega_l}+\varepsilon)\times(-\varrho_{\omega_l}-\varepsilon,0)$.)

If $\left(\frac{\bar{\partial}_+c(G)}{\sin^2\omega(G)},\ \frac{\bar{\partial}_-c(G)}{\sin^2\omega(G)}\right)\in\gamma_1$, then by the first equation of (2.35) we have $\bar{\partial}_+\left(\frac{\bar{\partial}_-c}{\sin^2\omega}\right)>0$ at the point G, which leads to a contradiction. If $\left(\frac{\bar{\partial}_+c(G)}{\sin^2\omega(G)},\ \frac{\bar{\partial}_-c(G)}{\sin^2\omega(G)}\right)\in\gamma_2$ then we have $-c\bar{\partial}_-\left(\frac{\bar{\partial}_+c}{\sin^2\omega}\right)<0$ at the point G, which leads to a contradiction. If $\left(\frac{\bar{\partial}_+c(G)}{\sin^2\omega(G)},\ \frac{\bar{\partial}_-c(G)}{\sin^2\omega(G)}\right)\in\gamma_3$, we let z be a smooth function defined on $\widetilde{GC_G}$ such that

$$\begin{cases}(-\bar{\partial}_-)z=\dfrac{\sin^2\omega}{2c\mu^2\cos^2\omega}\left(\dfrac{-\bar{\partial}_-c(G)}{\sin^2\omega(G)}-z\right)z\quad\text{along}\ \widetilde{GC_G};\\[2ex] z(C_G)=\dfrac{\bar{\partial}_+c(C_G)}{\sin^2\omega(C_G)}.\end{cases}\tag{4.81}$$

Then by integration we have $z(G)<\frac{-\bar{\partial}_-c(G)}{\sin^2\omega(G)}$, since $\frac{\bar{\partial}_+c(C_G)}{\sin^2\omega(C_G)}<\frac{-\bar{\partial}_-c(G)}{\sin^2\omega(G)}$. From the second equation of (2.35) we also have

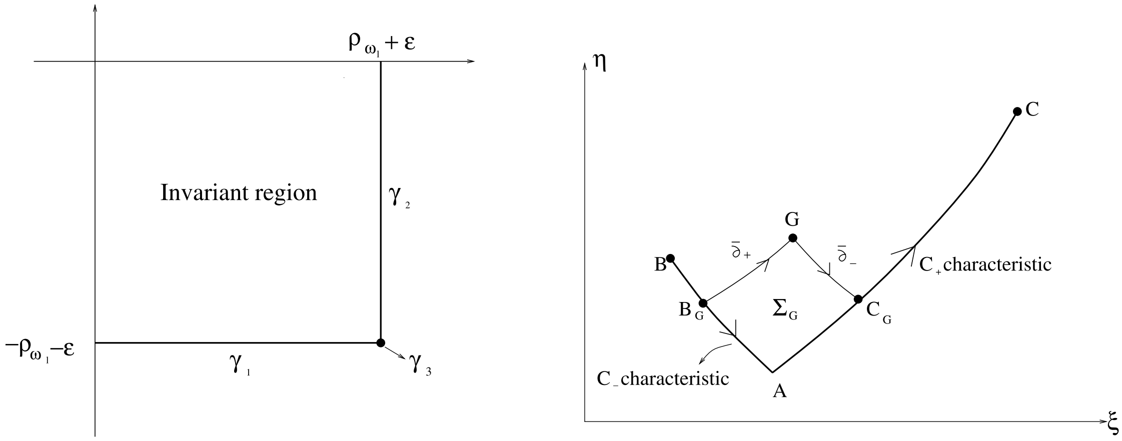

Fig. 7. Left: invariant region of $\left(\frac{\bar\partial_+ c}{\sin^2\omega}, \frac{\bar\partial_- c}{\sin^2\omega}\right)$; right: domain Σ_G.

$$
\begin{aligned}
-\bar\partial_-\Big(z-\frac{\bar\partial_+ c}{\sin^2\omega}\Big) &= \frac{\sin^2\omega}{2c\mu^2\cos^2\omega}\Big(\frac{-\bar\partial_- c(G)}{\sin^2\omega(G)}-\frac{\bar\partial_+ c}{\sin^2\omega}-z\Big)\Big(z-\frac{\bar\partial_+ c}{\sin^2\omega}\Big)\\
&\quad+\frac{\sin^2\omega}{2c\mu^2\cos^2\omega}\Big(\frac{\bar\partial_+ c}{\sin^2\omega}\Big)\underbrace{\Big(\frac{\bar\partial_- c}{\sin^2\omega}-\frac{\bar\partial_- c(G)}{\sin^2\omega(G)}\Big)}_{>0}-\Big(\frac{8\sin^2\omega}{\gamma-1}+2\Big)\Big(\frac{\bar\partial_+ c\bar\partial_- c}{\sin^2\omega}\Big)\\
&>\frac{\sin^2\omega}{2c\mu^2\cos^2\omega}\Big(\frac{-\bar\partial_- c(G)}{\sin^2\omega(G)}-\frac{\bar\partial_+ c}{\sin^2\omega}-z\Big)\Big(z-\frac{\bar\partial_+ c}{\sin^2\omega}\Big)
\end{aligned}
\tag{4.82}
$$

along $\widetilde{GC_G}$. Then by $\Big(z-\frac{\bar\partial_+ c}{\sin^2\omega}\Big)(C_G)=0$ we have $z(G)>\frac{\bar\partial_+ c(G)}{\sin^2\omega(G)}$ which leads to a contradiction. We then have Lemma 4.8. □

From (4.78) we immediately have

$$
\| Dc \|_{0;\Omega^G_{\omega_l}(\delta)} \le \frac{2\varrho_{\omega_l}}{\min\{\sin 2\omega_l, \sin 2\omega(A)\}},
\tag{4.83}
$$

since $\sin\beta\bar\partial_+ - \sin\alpha\bar\partial_- = -\sin 2\omega\partial_\xi$ and $\cos\beta\bar\partial_+ - \cos\alpha\bar\partial_- = \sin 2\omega\partial_\eta$. From (2.19)–(2.22), (4.83), and Lemma 4.6 we have the following gradient estimates for α and β:

$$
\| (D\alpha, D\beta) \|_{0;\Omega^G_{\omega_l}(\delta)} \le \frac{2}{\min\{\sin 2\omega_l, \sin 2\omega(A)\}}\left[\frac{\varrho_{\omega_l}(\tan^2\omega_l+\tan\omega_l)}{\mu^2 c^b(0,0)}+\frac{2}{c^b(0,0)}\right].
\tag{4.84}
$$

The gradient estimate for the variable φ is

$$
\| D\varphi \|_{0;\Omega^G_{\omega_l}(\delta)} \le \frac{c_M}{\sin\omega(A)}.
\tag{4.85}
$$

4.2.4. Global solution

Due to the degeneracy caused by pseudosonic, the classical approach to extend local solutions to global solutions does not work here. In what follows, we shall extend the local solution of the Goursat problem to global solution by solving many "small" Goursat problems in each extension step.

Assume that the Goursat problem admits a C^1 solution on $\Omega^G_{\omega_l}(\delta)$ where $\omega(A)<\omega_l<\frac{\pi}{2}$. Let $\widetilde{PQ}$ and $\widetilde{RP}$ be respectively a C_+ and a C_- characteristic curves in $\Omega^G_{\omega_l}(\delta)$; see Fig. 9 (left). We prescribe

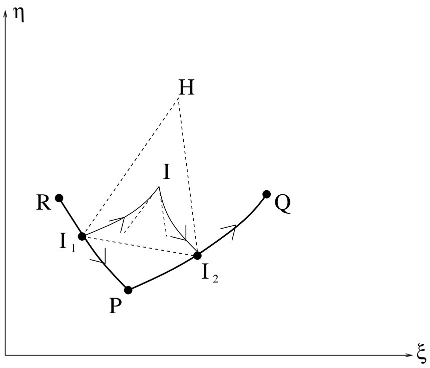

Fig. 8. Arc lengths of characteristics.

$$\begin{cases} (\alpha,\beta,\varphi) = \big(\alpha_{PQ},\ \beta_{PQ},\ \varphi_{PQ}\big) & \text{on } \widetilde{PQ}; \\ (\alpha,\beta,\varphi) = \big(\alpha_{RP},\ \beta_{RP},\ \varphi_{RP}\big) & \text{on } \widetilde{RP}, \end{cases} \tag{4.86}$$

where $\big(\alpha_{PQ},\ \beta_{PQ},\ \varphi_{PQ}\big)$ and $\big(\alpha_{RP},\ \beta_{RP},\ \varphi_{RP}\big)$ are the value of the solution on $\widetilde{PQ}$ and $\widetilde{RP}$, respectively. We then have the following theorem:

Lemma 4.9 *(Curved quadrilateral building block). When the arc lengths of $\widetilde{PQ}$ and $\widetilde{RP}$ are less than ε_0 where $\varepsilon_0 = \frac{\mu^2 c^b(0,0)\sin(\frac{\pi}{2}-\omega_l)}{3\varrho_{\omega_l}\tan^2(\frac{\pi}{4}+\frac{\omega_l}{2})}\cdot\big(\frac{\pi-2\omega_l}{4}\big)$, Goursat problem (2.18), (4.86) admits a global C^1 solution on a curved quadrilateral domain bounded by $\widetilde{PQ}$, $\widetilde{RP}$, $\widetilde{RT}$, and $\widetilde{QT}$, where $\widetilde{RT}$ is the C_+ characteristic passing through R, $\widetilde{QT}$ is the C_- characteristic passing through Q, and this solution satisfies $\omega_l \le \omega < \frac{\pi}{4}+\frac{\omega_l}{2}$.*

Proof. From Lemma 4.8 we have $\sup\limits_{\widetilde{RP}} \frac{\bar\partial_+ c}{\sin^2\omega} \le \varrho_{\omega_l}$ and $\sup\limits_{\widetilde{PQ}} \frac{-\bar\partial_- c}{\sin^2\omega} \le \varrho_{\omega_l}$, then by virtue of a similar method as that of Lemma 4.8 we have that the solution of Goursat problem (2.18), (4.86) satisfies

$$(\bar\partial_+ c,\ \bar\partial_- c) \in (0, \varrho_{\omega_l}] \times [-\varrho_{\omega_l}, 0). \tag{4.87}$$

Thus, we have

$$\begin{aligned} &0 < \bar\partial_+\alpha \le \frac{\varrho_{\omega_l}\tan^2\omega}{2\mu^2 c^b(0,0)},\ \ 0 < \bar\partial_-\beta \le \frac{\varrho_{\omega_l}\tan^2\omega}{2\mu^2 c^b(0,0)}, \\ &-\frac{\varrho_{\omega_l}\tan\omega}{\mu^2 c^b(0,0)} - \frac{2}{c^b(0,0)} < \bar\partial_+\beta < 0,\ \ -\frac{\varrho_{\omega_l}\tan\omega}{\mu^2 c^b(0,0)} < \bar\partial_-\alpha < 0. \end{aligned} \tag{4.88}$$

In what follows we shall show that when arc lengths of $\widetilde{PQ}$ and $\widetilde{RP}$ are less than ε_0, this solution satisfies

$$\begin{aligned} &\omega < \frac{\pi}{4}+\frac{\omega_l}{2}, \quad 0 < \bar\partial_+\alpha \le \frac{\varrho_{\omega_l}\tan^2(\frac{\pi}{4}+\frac{\omega_l}{2})}{2\mu^2 c^b(0,0)}, \quad 0 < \bar\partial_-\beta \le \frac{\varrho_{\omega_l}\tan^2(\frac{\pi}{4}+\frac{\omega_l}{2})}{2\mu^2 c^b(0,0)}, \\ &-\frac{\varrho_{\omega_l}\tan(\frac{\pi}{4}+\frac{\omega_l}{2})}{\mu^2 c^b(0,0)} - \frac{2}{c^b(0,0)} < \bar\partial_+\beta < 0, \quad -\frac{\varrho_{\omega_l}\tan(\frac{\pi}{4}+\frac{\omega_l}{2})}{\mu^2 c^b(0,0)} < \bar\partial_-\alpha < 0. \end{aligned} \tag{4.89}$$

We shall prove (4.89) by contradiction. Refer to Fig. 8. Suppose that there exists a point I such that $\omega(I) = \frac{\pi}{4}+\frac{\omega_l}{2}$. The C_+ characteristic curve passing through I intersects with $\widetilde{RP}$ at a point I_1, the C_- characteristic curve passing through I intersects with $\widetilde{PQ}$ at a point I_2. Through the point I_1 draw a straight line with the slope $\tan\alpha(I)$, through the point I_2 draw a straight line with the slope $\tan\beta(I)$, the two straight lines intersect at a point H. By estimate (4.70) we have $\widetilde{II_1}$ and $\widetilde{II_2}$ lie in the triangle $\triangle_{HI_1I_2}$

and the arc lengths of $\widetilde{II_1}$ and $\widetilde{II_2}$ are less than the sum of the arc lengths of $\overline{HI_1}$ and $\overline{HI_2}$ and $\overline{I_1I_2}$. Thus, the arc lengths of $\widetilde{II_1}$ and $\widetilde{II_2}$ are less than $\frac{6\varepsilon_0}{\sin(\frac{\pi}{2}-\omega_l)}$. Then, by (4.88) we have

$$\alpha(I) < \alpha(I_1) + \frac{\varrho_{\omega_l}\tan^2(\frac{\pi}{4}+\frac{\omega_l}{2})}{2\mu^2 c^b(0,0)} \cdot \frac{6\varepsilon_0}{\sin(\frac{\pi}{2}-\omega_l)} < \alpha(P) + \frac{\pi - 2\omega_l}{4} \tag{4.90}$$

and

$$\beta(I) > \beta(I_2) - \frac{\varrho_{\omega_l}\tan^2(\frac{\pi}{4}+\frac{\omega_l}{2})}{2\mu^2 c^b(0,0)} \cdot \frac{6\varepsilon_0}{\sin(\frac{\pi}{2}-\omega_l)} > \beta(P) - \frac{\pi - 2\omega_l}{4}. \tag{4.91}$$

Thus, we have $\omega(I) < \frac{\pi}{4} + \frac{\omega_l}{2}$, which leads to a contradiction.

From (4.89) we have that the solution satisfies

$$\begin{aligned}
&\alpha(P) \le \alpha \le \alpha(P) + \frac{\pi - 2\omega_l}{4}, \quad \beta(P) - \frac{\pi - 2\omega_l}{4} \le \beta \le \beta(P), \quad -\frac{c_M^2}{2\sin^2\omega(P)} - \frac{c_M^2}{\gamma - 1} < \varphi < 0, \\
&|\,(D\alpha, D\beta)\,| \le \frac{2}{\min\{\sin(\frac{\pi}{2}+\omega_l), \sin 2\omega(A)\}}\left[\frac{\varrho_{\omega_l}[\tan^2(\frac{\pi}{4}+\frac{\omega_l}{2}) + \tan(\frac{\pi}{4}+\frac{\omega_l}{2})]}{\mu^2 c^b(0,0)} + \frac{2}{c^b(0,0)}\right], \\
&|\,D\varphi\,| \le \frac{c_M}{\sin\omega(A)}.
\end{aligned} \tag{4.92}$$

Hence, by theory of global classical solutions for quasilinear hyperbolic equations (cf. Li [11]) we have this lemma. □

Theorem 4.10. *The Goursat problem (2.18), (4.65) admits a unique global C^1 solution on domain $\Omega^G(\delta)$ closed by $\widetilde{AC}$, $\widetilde{BA}$, $\widetilde{BC_\delta}$, and $\widetilde{CC_\delta}$. Here, $\widetilde{BC_\delta}$ is a C_+ characteristic curve passing through B and with $\omega(C_\delta) = \frac{\pi}{2}$, $\widetilde{CC_\delta}$ is a pseudosonic curve connecting C and C_δ; see Fig. 9 (right).*

Proof. Assume that the Goursat problem admits a C^1 solution on $\Omega^G_{\omega_l}(\delta)$ where $\omega(A) < \omega_l < \frac{\pi}{2}$, then by estimate (4.84) that the level curve $\omega(\xi,\eta) = \omega_l$ is a Lipschitz curve. Let $Q_0 = C_l,\ Q_1,\ Q_2,\ \cdots,\ Q_n = B_l$ be $n+1$ different points on the level curve $\omega = \omega_l$ in turn. The C_+ characteristic curve through Q_i intersects with the C_- characteristic curve through Q_{i+1} at a point P_i, where $i = 0, 1, \cdots, n-1$. Since $\bar{\partial}_+\omega > 0$ and $\bar{\partial}_-\omega < 0$, the level curve $\omega = \omega_l$ is a non-characteristic curve. Hence, $P_i \ne Q_i$ and $P_i \ne Q_{i+1}$ for any $i = 0, 1, \cdots, n-1$. Therefore, by using a similar method as that of Lemma 4.9 we know that there exists a sufficiently large n, such that the arc lengths of $\widetilde{P_iQ_i}$ and $\widetilde{P_iQ_{i+1}}$ are less than ε_0 for any $i = 0, 1, \cdots, n-1$. Thus, by Lemma 4.9 we know that, for any $i = 0, 1, \cdots, n-1$, Goursat problem for system (2.18) with $\widetilde{P_iQ_i}$ and $\widetilde{P_iQ_{i+1}}$ as characteristic boundaries admits a unique C^1 solution on curved quadrilateral domain bounded by $\widetilde{P_iQ_i}$, $\widetilde{P_iQ_{i+1}}$, $\widetilde{R_{i+1}Q_i}$, and $\widetilde{R_{i+1}Q_{i+1}}$, where $\widetilde{R_{i+1}Q_i}$ is the C_- characteristic curve passing through Q_i, $\widetilde{R_{i+1}Q_{i+1}}$ is the C_+ characteristic curve passing through Q_{i+1}.

Let $R_0 = C$, then for each $i = 0, 1, \cdots, n$, there exists a ω_i, $\omega_l < \omega_i < \frac{\pi}{2}$, such that Goursat problem for system (2.18) with $\widetilde{Q_iR_i}$ and $\widetilde{Q_iR_{i+1}}$ as the characteristic boundaries admits a unique C^1 solution on domain closed by $\widetilde{Q_iR_{i+1}}$, $\widetilde{Q_iR_i}$, and a level curve $\omega(\xi,\eta) = \omega_i$. Let $\omega_e = \min\{\omega_0, \omega_1, \cdots, \omega_n, \omega(R_1), \omega(R_2), \cdots, \omega(R_n)\}$. It is easy to see by $\bar{\partial}_+\omega > 0$ that $\omega_e > \omega_l$. Then, the local solution is extended from $\Omega^G_{\omega_l}(\delta)$ to $\Omega^G_{\omega_e}(\delta)$. Repeating these processes, we can construct the solution on $\Omega^G(\delta) = \bigcup_{\omega_l \in (\omega(A), \frac{\pi}{2})} \Omega^G_{\omega_l}(\delta)$.

From (4.72) and (4.73) we also have

$$\bar{\partial}_+\omega > \frac{\sin^2\omega}{c} > \frac{\sin^2\omega^b(0,0)}{c_M}, \tag{4.93}$$

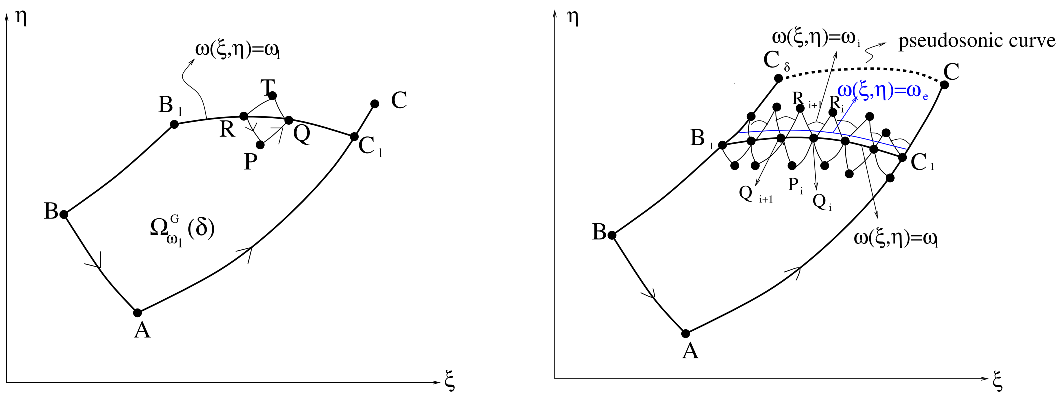

Fig. 9. Global classical solution to the Goursat problem.

which implies that the limit of the levels $\omega(\xi,\eta) = \omega_l$ as $\omega_l \to \frac{\pi}{2}$ exists. We then have Theorem 4.10. $\square$

4.3. Global C^1 solution to the centered wave bubble problem

According to Theorem 4.3 and Theorem 4.10, the solution on the domain $\Omega(\delta)$ closed by the C_+ characteristic curves $\widetilde{OC}$ and $\widetilde{OC_\delta}$ and sonic curve $\widetilde{CC_\delta}$ is obtained, and this solution belongs to $C^1\big(\overline{\Omega(\delta)} \setminus (0,0)\big)$ and satisfies

$$\bar{\partial}_+ c \,|_{\widetilde{OC_\delta}} > 0. \tag{4.94}$$

4.3.1. Regularity along C_+ characteristics

In order to extended the solution to $\Omega := \bigcup\limits_{\delta \in (0, \zeta_2 - \zeta_1)} \Omega(\delta)$, we need the following estimate.

Lemma 4.11. *Assume that the centered wave problem admits a C^1 solution on $\Omega(\iota)$, $0 < \iota < \zeta_2 - \zeta_1$. Let G be an arbitrary point on $\widetilde{OC_\iota}$, where $\widetilde{OC_\iota}$ is the C_+ characteristic curve passing through O and with the slope $\zeta_1 + \iota$ at O, then we have*

$$0 < \bar{\partial}_+ c(G) \le \frac{c_M}{c^b(0,0)} e^{\overline{\frac{2}{2\mu^2 \cos^2 \omega(G)}}} \cdot \sup_{\widetilde{AC_G}} \frac{\bar{\partial}_+ c}{\sin^2 \omega}, \tag{4.95}$$

where C_G is the point on $\widetilde{OC}$ with $\omega(C_G) = \omega(G)$.

Proof. Sine $\bar{\partial}_- \omega < 0$, the C_- characteristic curve passing through G intersects with $\widetilde{OC_G}$ at some point which is denoted by H. Then, by the second equation of (2.35) we have along $\widetilde{GH}$,

$$-\bar{\partial}_- \left(\frac{\bar{\partial}_+ c}{\sin^2 \omega} \right) < -\frac{1}{2\mu^2 \cos^2 \omega} \left(\frac{\bar{\partial}_+ c}{\sin^2 \omega} \right) \frac{\bar{\partial}_- c}{c}. \tag{4.96}$$

Integrating this along $\widetilde{HG}$ from H to G and noting that $\omega < \omega(G)$ along $\widetilde{GH}$, we get

$$\frac{\bar{\partial}_+ c(G)}{\sin^2 \omega(G)} < \frac{c(G)}{c(H)} \frac{\bar{\partial}_+ c(H)}{\sin^2 \omega(H)} e^{\overline{\frac{1}{2\mu^2 \cos^2 \omega(G)}}} \le \frac{c_M}{c^b(0,0)} e^{\overline{\frac{2}{2\mu^2 \cos^2 \omega(G)}}} \cdot \sup_{\widetilde{AC_G}} \frac{\bar{\partial}_+ c}{\sin^2 \omega}. \tag{4.97}$$

We then have this lemma. $\square$

4.3.2. Global C^1 solution to the centered wave bubble problem

According to Theorem 4.3 and Theorem 4.10, we know that there exists a $\delta > 0$, δ depends only on m and ϑ, such that the centered wave bubble problem admits a solution on the domain $\Omega(\delta)$. We then consider a similar centered wave problem with $\widetilde{OC_\delta}$ as the C_+ characteristic boundary. By Lemma 4.11, we know that there exists a $\delta' > 0$, such that the solution of the centered wave problem can be extend from $\Omega(\delta)$ to $\Omega(\delta + \delta')$. Repeating above process, we know that the solution can be extended to Ω, and by (4.93) we know that Ω is bounded by $\widetilde{OC}$ and a pseudosonic curve connection O and C.

Remark 4.12. In a recent paper [6], we study the question of existence of centered expansion fans for 2D steady Guderley Mach reflection configurations. Due to lake of an estimate similar to (4.93) the sonic boundary of the centered wave problem in 2D steady case may go to infinity.

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